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***Investigating FOXP1-Regulated Gene Networks in
Schizophrenia Pathogenesis***

Examination candidate's name: Brian Oghene

ID number: 21422876

Academic supervisor: Derek Morris

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BI453 Research Report

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Student name: Brian Oghene

ID number: 21422876

Report title: Investigating FOXP1-Regulated Gene Networks in Schizophrenia Pathogenesis

Supervisor: Derek Morris

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List of Abbreviations

<i>Abbreviation</i>	<i>Full Term</i>
<i>cKO</i>	<i>Conditional Knockout</i>
<i>DEG</i>	<i>Differentially Expressed Gene</i>
<i>GEO</i>	<i>Gene Expression Omnibus</i>
<i>GWAS</i>	<i>Genome-Wide Association Study</i>
<i>LD</i>	<i>Linkage Disequilibrium</i>
<i>MAGMA</i>	<i>Multi-marker Analysis of GenoMic Annotation</i>
<i>NCBI</i>	<i>National Centre for Biotechnology Information</i>
<i>RNA-seq</i>	<i>RNA Sequencing</i>
<i>SRA</i>	<i>Sequence Read Archive</i>
<i>SIFT</i>	<i>Sorting Intolerant from Tolerant</i>
<i>SNP</i>	<i>Single Nucleotide Polymorphism</i>
<i>SynGO</i>	<i>Synaptic Gene Ontology</i>
<i>TPM</i>	<i>Transcripts Per Million</i>
<i>UTR</i>	<i>Untranslated Region</i>
<i>VEP</i>	<i>Variant Effect Predictor</i>

1. Lay Abstract

Schizophrenia is a complex brain disorder that affects how a person thinks, feels and behaves. This study used advanced computer techniques to investigate the genetic roots of the condition by examining both mouse and human genetic data. This study helped to discover several key genes that might play a role in schizophrenia, with special attention to how these genes affect communication between brain cells. The research found that certain genes particularly one called *FOXP1*, could be important in understanding the disorder. By mapping how these genes work it was possible to gain insights into why schizophrenia develops and how it impacts brain function. This approach provides a promising pathway for future research that could potentially lead to better treatments by targeting the specific genetic mechanisms underlying the condition.

2. Scientific Abstract

Schizophrenia is a complex neurodevelopmental disorder characterized by a complex genetic etiology and synaptic dysfunction. This study utilized bioinformatics approaches to investigate the molecular mechanisms underlying schizophrenia by examining *FOXP1*, a transcription factor associated with the disorder, its regulated gene networks, and their intersection with schizophrenia risk loci. Utilising RNA sequencing data from *FOXP1* conditional knock-out (cKO) mouse models, a multi-step computational pipeline was employed, integrating differential gene expression analysis, genome-wide association (GWAS) data, and advanced functional annotation tools. Transcriptomic data were processed using SRA Toolkit, FastQC, Trimmomatic, and Salmon, with differential expression analysis conducted via DESeq2. Gene-based association analysis was performed using MAGMA which gave a p-value of 0.047 showing that gene set was enriched. This was complemented by comprehensive annotation through SynGO, LocusZoom, HaploReg, and Ensembl Variant Effect Predictor. A total of 349 differentially expressed genes (DEGs) were identified, with 11 genes showing significant genetic association with schizophrenia after stringent Bonferroni correction. SynGO analysis revealed enrichment of synaptic processes, with significant overrepresentation of DEGs in postsynaptic specialization and synaptic transmission pathways. Key candidate genes including *ZNF804A* and *NEGR1* demonstrated strong genetic associations and potential functional relevance. This research provides insights into the molecular mechanisms of schizophrenia, highlighting the importance of synaptic gene networks and transcriptional regulation. By bridging mouse model research with human genetic analysis, the study offers a comprehensive framework for understanding the complex genetic architecture of schizophrenia and provides promising avenues for future therapeutic targeting.

3. Introduction

Schizophrenia is a complex and debilitating psychiatric disorder that affects approximately 1% of the global population [1]. It is characterised by a profound disruption in cognition, emotion and perception manifesting in a wide array of symptoms that significantly impair an individual's ability to function in society.[1] Schizophrenia typically manifests in late adolescence or early adulthood, a crucial stage for cognitive growth and social development. Around 8% of cases are diagnosed before age 18, while approximately 18% of individuals experience their first symptoms during this period [2]. As such, the disorder has profound implications for the affected individual's life trajectory, and it presents a major challenge to healthcare systems worldwide. Synapses, the junctions between neurons, are critical for transmitting signals across the nervous system through neurotransmitters, and disruptions in synaptic function have been implicated in the pathophysiology of schizophrenia [3]. Understanding schizophrenia requires a multifaceted approach considering both the genetic and environmental factors that contribute to its development as well as the molecular mechanisms underlying its pathophysiology.

3.1 Symptoms of Schizophrenia

Schizophrenia is classically divided into three main symptom clusters - positive, negative and cognitive symptoms. Positive symptoms refer to the presence of abnormal experiences or behaviours such as delusions, hallucinations and disorganised thinking [1]. Delusions are false beliefs that are strongly held despite evidence to the contrary, while hallucinations are sensory experiences that occur in the absence of external stimuli. The most common type of hallucinations in schizophrenia are auditory, where individuals hear voices that others do not. Disorganised thinking, often manifesting as incoherent speech, is another hallmark of the disorder and reflects a disruption in cognitive processes.

Negative symptoms refer to the absence or reduction of normal functioning [4]. These include diminished emotional expression, reduced motivation and social withdrawal which contribute to difficulties in personal relationships and social integration. The cognitive symptoms of schizophrenia involve impairments in memory, attention and executive functioning. These cognitive deficits are among the most debilitating aspects of schizophrenia and are strongly associated with poor functional outcomes. While the specific combination and severity of symptoms can vary among individuals, the impact on daily life is often profound. Individuals with schizophrenia may struggle to maintain employment, attend school or live independently. The disorder is also associated with a significantly increased risk of comorbid conditions such as depression, anxiety and substance abuse [2].

3.2 Environmental Factors

The development of schizophrenia is thought to be influenced by a complex interplay of genetic predispositions and environmental factors. Research suggests that environmental stressors, particularly during critical periods of neurodevelopment can contribute to the onset and progression of schizophrenia in genetically predisposed individuals [5]. Among the most significant environmental factors are prenatal exposures to infections, malnutrition and stress.

Prenatal stress is one of the most consistently studied risk factors for schizophrenia. Studies have shown that prenatal and perinatal complications may contribute to an increased risk of schizophrenia, with obstetric factors generally showing odds ratios between 1 and 2 for psychotic disorders [6]. Infections during pregnancy particularly those affecting the central nervous system have been linked to a heightened risk of schizophrenia. For example, exposure to the influenza virus *in utero* has been associated with an increased risk of developing schizophrenia later in life [5].

Growing up or residing in an urban environment has consistently been linked to a heightened risk of schizophrenia and other psychotic disorders. Research indicates that individuals raised in cities, particularly those from lower socioeconomic backgrounds are more susceptible to developing schizophrenia compared to those from rural areas [5]. This increased risk is attributed to various environmental stressors associated with urban living, such as social stress, pollution and adverse childhood experiences.

A comprehensive meta-analysis, which included 47,087 cases of psychosis reported that individuals living in urban environments had an approximately 2.39 times higher risk of developing psychosis compared to those in rural areas [5]. Moreover, studies suggest that relocating from a rural to an urban area during childhood can double the risk of schizophrenia. Additionally, the longer a child spends in an urban setting the greater the likelihood of developing the disorder [5].

3.3 Genetic Factors and *FOXP1* Gene

Schizophrenia has a strong genetic component, with heritability estimates ranging from 60% to 80% [7]. This suggests that genetic factors play a significant role in the development of the disorder, although environmental influences are also crucial. Several genes have been implicated in schizophrenia many of which are involved in neural development, neurotransmission, and synaptic plasticity. Among the most studied genes is the dopamine receptor D2 gene (*DRD2*), which is involved in the regulation of dopamine, a neurotransmitter that is implicated in many of the symptoms of schizophrenia, particularly the positive symptom [7]. However, schizophrenia is considered a polygenic disorder meaning that multiple genes each with a small effect contribute to its development [7]. This has been determined through genome-wide association studies (GWAS), which have identified hundreds of genetic variants associated with schizophrenia risk [8].

One schizophrenia risk gene that has garnered attention in recent research is Forkhead Box P1 (*FOXP1*), a member of the forkhead box (FOX) family of transcription factors. *FOXP1* plays a crucial role in neuronal development, including the regulation of synaptic connections and neural plasticity [9]. Alterations in the expression or function of *FOXP1* have been associated with various neurodevelopmental disorders, including schizophrenia. Studies have shown that mutations or deletions of *FOXP1* can lead to disruptions in brain development, particularly in areas associated with language, memory and social cognition, which are often impaired in schizophrenia [9]. Dysregulation of *FOXP1* may contribute to the structural and functional abnormalities observed in the brains of individuals with schizophrenia, particularly in the prefrontal cortex and hippocampus areas which are critical for executive function and memory [9].

3.4 Aim of this study

Recent studies of *FOXP1* conditional knock-out (*cKO*) mice have demonstrated that loss of this transcription factor in forebrain neurons leads to synaptic gene dysregulation and behavioral phenotypes relevant to neurodevelopmental disorders [9]. Building on these findings, this study aims to investigate whether *FOXP1*-regulated gene networks contribute to schizophrenia-relevant synaptic and behavioral deficits by examining overlap between *FOXP1*-regulated genes and schizophrenia risk loci, characterizing synaptic function in *FOXP1* *cKO* mice, and identifying convergent molecular pathways. I plan to achieve these objectives through bioinformatic integration of *FOXP1* KO transcriptomic data with schizophrenia GWAS results, and pathway enrichment analyses. This approach will clarify whether *FOXP1* dysfunction represents a shared mechanism across neurodevelopmental disorders and provide insights into synaptic pathophysiology in schizophrenia.

4.Methods

This study integrates RNA Sequencing (RNA-seq) data analysis and GWAS data to investigate gene expression changes and genetic associations related to schizophrenia. The RNA-seq data were processed using bioinformatics tools for quality control, transcript quantification, and differential gene expression analysis. The differential expression results were then mapped to human genes and compared with GWAS-derived gene-based association results obtained through MAGMA. This multi-step approach enables the identification of genes with both expression alterations and genetic associations with schizophrenia, providing insights into potential molecular mechanisms underlying the disorder.

4.1 Table of Tools of Bioinformatic Tools Used

<i>Tool</i>	<i>Function</i>
SRA Toolkit (fastq-dump)	Downloads raw sequencing data from the Sequence Read Archive (SRA) and converts it into FASTQ format for downstream processing. [10]
FastQC	Performs quality control on raw sequencing reads by analysing sequence quality scores, GC content, sequence duplication levels, overrepresented sequences and adapter contamination. Generates graphical reports.
Trimmomatic	Preprocesses raw reads by removing adapter sequences, low-quality bases, and short reads. Uses a sliding window approach to trim reads based on quality thresholds and ensures high-quality data for downstream analysis. [11]
Salmon	Rapidly quantifies transcript abundance from RNA-Seq data using a lightweight, alignment-free, quasi-mapping approach. Incorporates sequence-specific and GC-content bias correction for accurate quantification. [12]
DESeq2	Performs differential gene expression analysis by modeling RNA-Seq count data using negative binomial distribution. Provides statistical methods for identifying significantly differentially expressed genes (DEGs) between conditions.[13]

4.2 Data Availability

The RNA-seq data used in this study are publicly available in the Gene Expression Omnibus (GEO) (<https://www.ncbi.nlm.nih.gov/geo>) under accession number GSE97181. This dataset includes sequencing data from control and cKO mouse samples specifically from the hippocampus. The dataset comprises 8 samples: 4 control and 4 cKO samples, enabling a robust comparison of gene expression profiles between these two conditions.

4.3 Raw Data Acquisition and Preprocessing

RNA-seq data were retrieved from the National Centre for Biotechnology Information (NCBI) Sequence Read Archive (SRA) (<https://www.ncbi.nlm.nih.gov/sra>) using SRA Toolkit. The FASTQ files corresponding to multiple sequencing runs were downloaded and stored in compressed format for efficient storage and processing. The following command was used to retrieve and preprocess the raw sequencing data:

```
(fastq-dump --split-3 --gzip)
```

The quality of the raw sequencing reads was assessed using FastQC, which provided an overview of various quality metrics, including base quality scores, adapter contamination, and sequence duplication levels. To analyse all FASTQ files in the dataset, the following command was executed:

```
for i in *.fastq.gz; do fastqc $i; done
```

To ensure high-quality reads for downstream analysis, preprocessing steps were conducted using Trimmomatic [11]. This involved removing adapter sequences, trimming low-quality bases and filtering out reads below a defined length threshold. The output of this step was a set of cleaned and quality-filtered FASTQ files. The following bash script was used to preprocess all FASTQ files in the dataset:

```
for file in *.fastq; do
    BASENAME=$(basename "$file" .fastq)
    OUTPUT_FILE="${BASENAME}.trimmed.fastq"
    trimmomatic SE -phred33 "$file" "$OUTPUT_FILE"
    ILLUMINACLIP:"$ADAPTER_FILE":2:30:10 LEADING:3 TRAILING:3
    SLIDINGWINDOW:4:15 MINLEN:36
done
```

4.4 Transcript Quantification

To quantify transcript abundance, the Salmon tool was employed [12]. A transcriptome index was first generated to facilitate rapid alignment-free quantification. The cleaned reads were then pseudo-aligned to the reference transcriptome and transcript-level abundance estimates were obtained. The quantification process was performed with sequence-based bias correction and validation mappings to enhance accuracy. The output files contained transcript expression levels in terms of Transcripts Per Million (TPM) and raw read counts, which were used for subsequent differential gene expression analysis. The quantification process was performed using the following Salmon command:

```
salmon quant -i salmon_index -l A -r  
${SAMPLE}.trimmed.fastq.gz --validateMappings -o  
salmon_out/${SAMPLE} --threads 3
```

4.5 Differential Gene Expression Analysis

The quantification results from Salmon were analyzed using DESeq2 to identify differentially expressed genes (DEGs) between the experimental conditions [13]. Transcript-level quantification data were imported using tximport and summarized to the gene level using a transcript-to-gene mapping file generated from biomaRt. The raw counts were normalized within DESeq2 to account for differences in sequencing depth and RNA composition. A negative binomial distribution model was applied to estimate gene-wise dispersion and test for differential expression. Genes were considered significantly differentially expressed if they met the threshold criteria of an adjusted p-value. The results including log2 fold changes and adjusted p-values were saved to a CSV file for downstream analysis. Additionally, an MA plot was generated to visualize the relationship between gene expression levels and log2 fold changes.

4.6 Cross-Species Gene Mapping Using biomaRt

For cross-species comparison, mouse genes were mapped to their human homologs using biomaRt. The mouse gene symbols and Ensembl gene IDs were retrieved from the Ensembl database (<https://www.ensembl.org/index.html>) and the corresponding human gene symbols were fetched using the `hsapiens_homolog_associated_gene_name` attribute. The resulting mouse-to-human gene mapping was saved to a CSV file for downstream functional annotation and comparative analysis. The following R script was used for gene mapping:

```
mouse_mapping <- getBM(attributes = c("mgi_symbol",
  "ensembl_gene_id"),

  mart = ensembl)

human_mapping <- getBM(attributes = c("ensembl_gene_id",
  "hsapiens_homolog_associated_gene_name"),

  mart = ensembl)

gene_mapping <- merge(mouse_mapping, human_mapping, by =
  "ensembl_gene_id", all.x = TRUE)

colnames(gene_mapping)[colnames(gene_mapping) == "mgi_symbol"]
<- "mouse_gene_symbol"

colnames(gene_mapping)[colnames(gene_mapping) ==
  "hsapiens_homolog_associated_gene_name"] <-
  "human_gene_symbol"

write.csv(gene_mapping,
  "C:/Users/boghe/Downloads/salmon_out/mouse_to_human_gene_mappi
ng.csv", row.names = FALSE)
```

4.7 Volcano Plot Visualization

To visually represent the results of the differential gene expression analysis, a volcano plot was generated using ggplot2 in R. The log2 fold change values were plotted against the negative log10-transformed adjusted p-values to illustrate the significance and magnitude of expression changes. Genes classified as differentially expressed were highlighted in red, while non-significant genes were displayed in grey. Threshold lines were included to indicate cutoff values for significance.

The most significant DEGs were ranked by adjusted p-value, and their annotations were added to the plot using ggrepel to facilitate identification. The resulting volcano plot provided a clear visualization of the distribution of DEGs and was exported as a high-resolution PNG file for further analysis and reporting. The following R script was used for volcano plot analysis:

```
library(ggrepel) library(ggplot2)

res_df <- as.data.frame(res)

res_df$Significance <- "Not Significant"

res_df$Significance[res_df$padj < 0.05 &
abs(res_df$log2FoldChange) > 1] <- "Significant"

volcano_plot <- ggplot(res_df, aes(x = log2FoldChange, y = -
log10(padj), color = Significance)) + geom_point(alpha = 0.5,
size = 1) + scale_color_manual(values = c("Significant" =
"red", "Not Significant" = "grey")) + geom_vline(xintercept =
c(-1, 1), linetype = "dashed", color = "black", linewidth =
0.4, alpha = 0.5) + geom_hline(yintercept = -log10(0.05),
linetype = "dashed", color = "black", linewidth = 0.4, alpha =
0.5) + labs(title = "Volcano Plot: DEGs (KO vs CTL)", x =
"Log2 Fold Change", y = "-Log10 Adjusted p-value") +
theme_minimal()

top_genes <- res_df[order(res_df$padj, na.last = NA),][1:10,]

volcano_plot <- volcano_plot + geom_text_repel(data =
top_genes, aes(label = human_gene_symbol), size = 3)

print(volcano_plot)
```

5. Gene-Based and Gene-Set Analysis Using MAGMA

To identify genes and biological pathways associated with schizophrenia, Multi-marker Analysis of GenoMic Annotation (MAGMA) was employed. MAGMA aggregates the effects of multiple Single Nucleotide Polymorphism (SNPs) within genes, providing a powerful approach to interpret GWAS results in a biologically meaningful context [14]. The analysis consisted of three main steps: SNP-to-gene annotation, gene-based analysis, and gene-set analysis.

The first step involved annotating SNPs to genes using genomic coordinates. The 1000 Genomes Project European population data and the NCBI Build 37 gene location file were used to map SNPs to their respective genes. The following command was executed:

```
./magma --annotate --snp-loc g1000_eur.bim --gene-loc  
NCBI37.3.gene.loc --out NCBI37_eur
```

Gene-based association testing was performed to identify genes significantly associated with schizophrenia. The GWAS summary statistics from the most recent schizophrenia GWAS were used, with a sample size of 320,404 individuals [8]. The following command was executed:

```
./magma --bfile g1000_eur --pval PGC3_SCZ_wave3_public.tsv  
N=320404 --gene-annot NCBI37_eur.genes.annot --out Pergene_SCZ
```

Gene-set analysis was conducted to test for enrichment of predefined biological pathways or functional gene sets in the schizophrenia GWAS data. The gene-based association results (Pergene_SCZ.genes.raw) and a gene-set annotation file were used as inputs.

5.1 Gene Symbol Conversion

Human gene symbols from the file geneset_magma.txt were converted to Entrez IDs using the biomaRt package in R. Converting gene symbols to Entrez IDs ensures that the input data is compatible with MAGMA's requirements, enabling accurate mapping of genes to their respective SNPs, pathways, or functional annotations.

The following code was executed:

```
setwd('C:/Users/boghe/OneDrive/Documents')
genes <- readLines("geneset_magma.txt")

require("biomaRt")
human <- useMart(biomart = "ENSEMBL_MART_ENSEMBL",
                 dataset = "hsapiens_gene_ensembl",
                 host = "https://grch37.ensembl.org")

genesV2 <- getLDS(attributes = c("hgnc_symbol"), filters =
"hgnc_symbol", values = genes,
                 mart = human, attributesL =
c("hgnc_symbol","entrezgene_id"), martL = human, uniqueRows=T)
humanx <- unique(genesV2)

write.table(humanx, "Gene_Set_Converted.txt", quote = F,
row.names = F, sep = "\t")
```

The Entrez IDs were formatted into a gene-set file compatible with MAGMA. The following code was executed:

```
magma_set <- t(humanx$NCBI.gene..formerly.Entrezgene..ID)
row.names(magma_set) <- "Gene_Set"

write.table(magma_set, quote = F, sep = "\t", file =
"Gene_Set_Tabs.txt", col.names = F)
```


Gene-set enrichment analysis was conducted to identify biologically relevant pathways and functional gene sets associated with schizophrenia. The analysis utilized gene-based association results generated by MAGMA (Pergene_SCZ.genes.raw) and a predefined gene-set annotation file (Gene_Set_Tabs.txt). The following command was executed to perform the analysis:

```
./magma --gene-results Pergene_SCZ.genes.raw --set-annot  
Gene_Set_Tabs.txt --out Gene_set_SCZ_MAGMA
```

This command evaluated the enrichment of gene sets within the schizophrenia GWAS dataset, employing a statistical framework to assess whether specific biological pathways or functional categories were overrepresented among genes associated with schizophrenia. The output files, prefixed as Gene_set_SCZ_MAGMA, included p-values and enrichment statistics for each gene set, facilitating the identification of pathways with significant genetic contributions to schizophrenia. This analysis provided a functional interpretation of the GWAS results, linking genetic associations to underlying biological mechanisms.

5.2 Identification of Highly Enriched DEGs

To integrate the results of DEG analysis from RNA-seq and gene-based association analysis from MAGMA, a systematic approach was employed. First, the DEG results, comprising gene identifiers, log2 fold changes and adjusted p-values, were combined with a mouse-to-human gene mapping file to annotate mouse genes with their corresponding human homologs. This step ensured cross-species comparability and facilitated downstream analysis in a human context.

Next, the annotated DEG results were merged with the MAGMA gene-based association results using human gene symbols as the common identifier. The merged dataset included per-gene p-values from MAGMA, along with additional gene-level statistics such as the number of SNPs and z-scores. To account for multiple testing, a Bonferroni correction was applied to the MAGMA p-values, establishing a stringent significance threshold of 2.76E-06. The final dataset was sorted by MAGMA p-value to prioritise the most significant genes.

6. Analysis of DEGs

6.1 SynGO

SynGO (<https://www.syngoportal.org/>) is a specialized gene ontology database focused on synaptic genes and their functional roles [15]. In this study, SynGO was used to annotate genes associated with schizophrenia, categorizing them into synaptic roles such as presynaptic or postsynaptic functions, synaptic vesicle cycling and signalling pathways. Functional enrichment plots and network diagrams were created to visualize the overrepresentation of synaptic processes among candidate genes, helping to prioritize those most relevant to synaptic biology. These results were integrated with LocusZoom and MAGMA outputs, which refined the list of candidate genes for further validation in the context of schizophrenia.

6.2 LocusZoom

LocusZoom (<https://my.locuszoom.org/>) generates regional association plots for genetic loci using GWAS results and visualizes the relationship between SNPs and their association signals [16]. In this study, LocusZoom was used to generate regional plots for schizophrenia-related SNPs which were categorized into high, moderate and low priority groups based on their association strength and linkage disequilibrium (LD) patterns. The refined list of SNPs and genes from LocusZoom was used in subsequent analyses with HaploReg and VEP for further functional annotation.

6.3 HaploReg

HaploReg (<https://pubs.broadinstitute.org/mammals/haploreg/haploreg.php>) annotates coding and non-coding variants and their regulatory impacts [17]. In this study, HaploReg was used to assess SNPs identified in LocusZoom, focusing on their locations within genomic regions (intronic, UTR, or exonic) and their potential regulatory roles. The integration of HaploReg results with LocusZoom and SynGO annotations allowed for prioritization of SNPs with the greatest functional relevance to synaptic functions. This refined list of candidate SNPs was then subjected to further analysis using Ensembl VEP to predict their functional consequences.

6.4 UCSC Genome Browser

The UCSC Genome Browser (<https://genome.ucsc.edu/>) provides a visual platform for exploring genomic regions, annotations, and experimental data [18]. This tool was used to examine SNPs identified in HaploReg, checking their location in gene structures and assessing their functional impact. The browser allowed for the visualization of SNPs within the context of regulatory regions and chromatin states which were cross-referenced with data from HaploReg and LocusZoom to refine the biological interpretation of their potential roles in schizophrenia.

6.5 Ensembl Variant Effect Predictor (VEP)

Ensembl VEP (<https://www.ensembl.org/info/docs/tools/vep/index.html>) annotates the functional effects of genetic variants [19]. VEP was used to predict the consequences of SNPs identified through HaploReg, particularly missense and Untranslated Region (UTR) variants. Sorting Intolerant from Tolerant (SIFT) scores were used to assess the potential impact of missense mutations on protein function and UTR variants were analysed for their possible effects on gene expression. The integration of VEP annotations with HaploReg, LocusZoom, and SynGO results helped prioritize SNPs with the highest functional relevance, guiding the selection of candidate variants for further experimental validation and enhancing the understanding of their biological roles in schizophrenia.

7. Results of Full DEG dataset

7.1 Volcano plot

A total of 28,866 genes were analysed, with 349 genes identified as DEGs (Supplementary Table 1). The volcano plot (Fig. 1) was generated to visualize the results of the DEG analysis prior to MAGMA analysis, highlighting the relationship between the magnitude of gene expression changes and their statistical significance. The plot provides a clear representation of the distribution of DEGs between the experimental conditions (e.g., knockout vs. control). The volcano plot reveals a clear separation of genes into three categories: Significantly Upregulated Genes, Significantly Downregulated Genes and Non-Significant Genes. Genes such as FYB2, SLC9A2, and HOMER2 are highlighted as key candidates due to their strong association signals and significant expression changes. These genes may play important roles in the biological processes underlying the experimental condition.

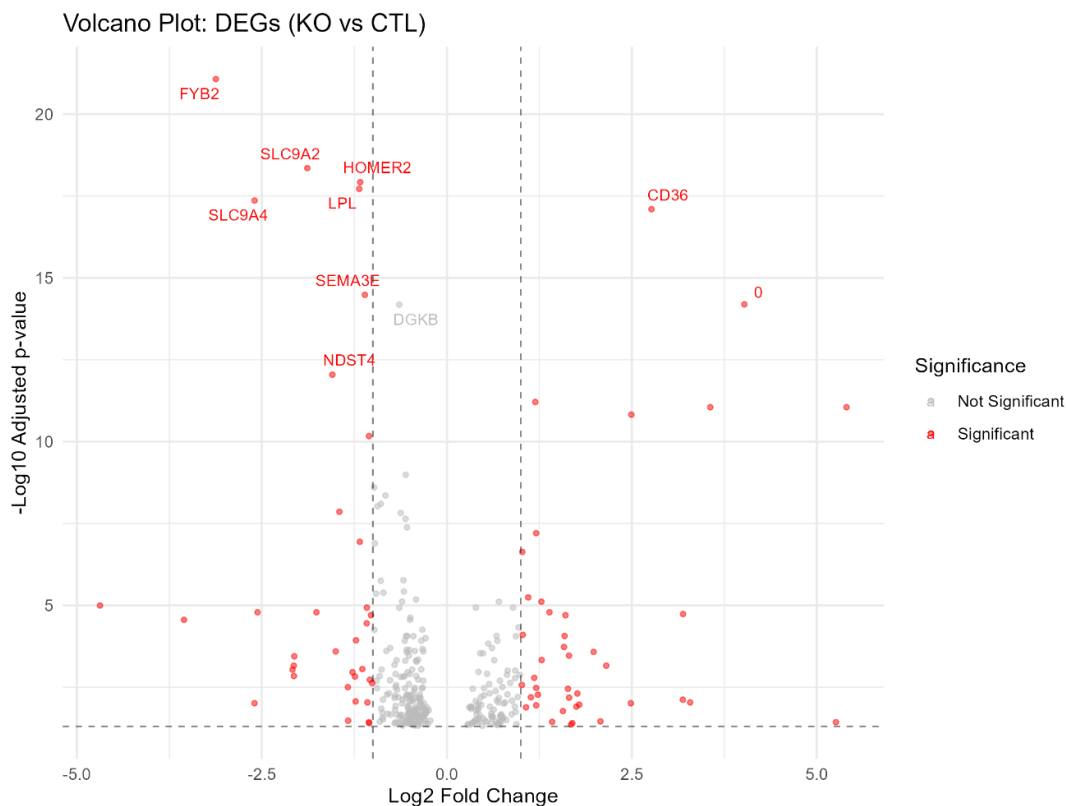


Figure 1. Volcano Plot of Differential Gene Expression Analysis. The volcano plot displays the relationship between log2 fold change and negative log10 adjusted p-value for genes analyzed in the differential expression study. Red points represent significantly differentially expressed genes, while grey points represent non-significant genes. The most significant DEGs are labeled.

7.2 MA PLOT

The MA plot (Fig.2) provides a global overview of the differential expression analysis, highlighting genes with significant changes in expression between the experimental conditions. Genes with high average expression levels and large log2 fold changes are likely to be biologically relevant and may play key roles in the observed phenotype. The plot also helps to identify potential biases or artifacts in the data, such as uneven distribution of expression changes or technical noise.

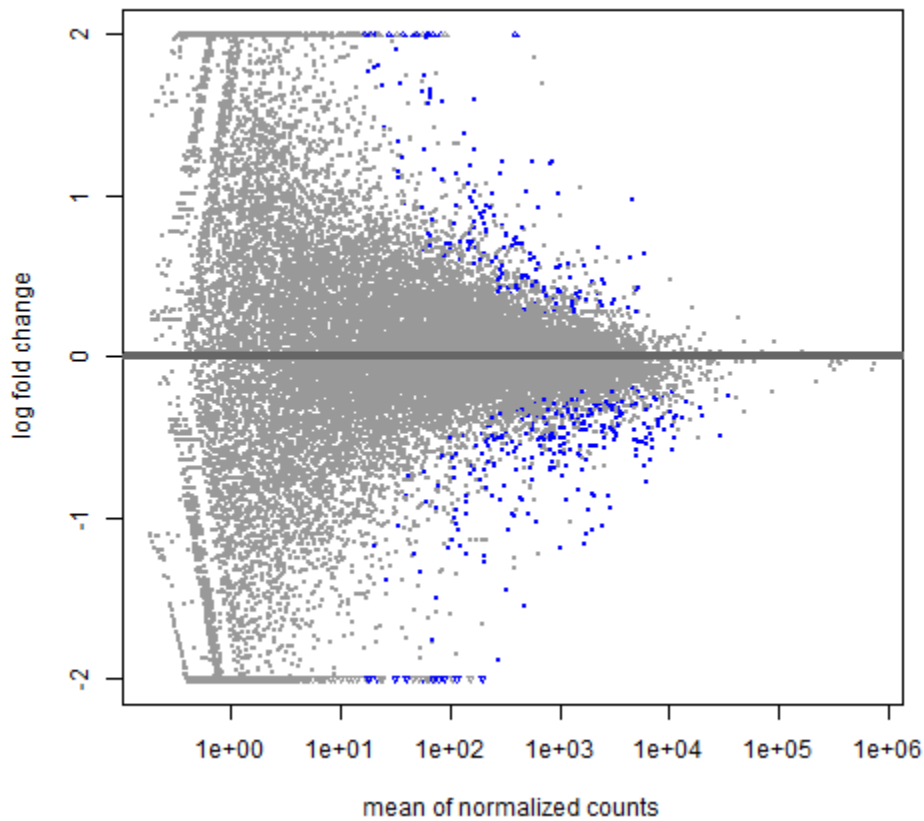


Figure 2. MA Plot of Differential Gene Expression Analysis. The MA plot displays the relationship between the average expression level of genes and the log2 fold change between the experimental conditions. Genes with significant expression changes are highlighted in blue, with upregulation and downregulation indicated above and below the line.

7.3 SYNGO Analysis

The SynGO analysis performed on the DEGs identified in this study revealed a strong enrichment for synaptic components and processes, underscoring the relevance of synaptic dysfunction in schizophrenia. The DEGs were highly associated with key synaptic structures (Fig.3), particularly the postsynapse and presynapse, as evidenced by the significant enrichment of terms such as synapse ($p = 2.43\text{e-}15$), postsynapse ($p = 3.97\text{e-}15$), and postsynaptic specialization ($p = 1.86\text{e-}09$). These findings align with the well-established role of postsynaptic abnormalities, such as altered glutamate receptor signaling in schizophrenia. Additionally, the DEGs were enriched for biological processes critical to synaptic function, including processes in the synapse ($p = 1.12\text{e-}12$), synapse organisation ($p = 5.77\text{e-}06$), and chemical synaptic transmission ($p = 0.00221$), highlighting their involvement in synaptic signaling, organisation, and neurotransmitter regulation. The enrichment of presynaptic terms, such as presynapse ($p = 1.58\text{e-}05$) and presynaptic membrane ($p = 6.54\text{e-}05$), further suggest disruptions in presynaptic mechanisms, such as neurotransmitter release and vesicle cycling. These results provide biological validation of the significant DEGs linking them to synaptic pathways implicated in schizophrenia. (Full list of results in Appendix- Github)

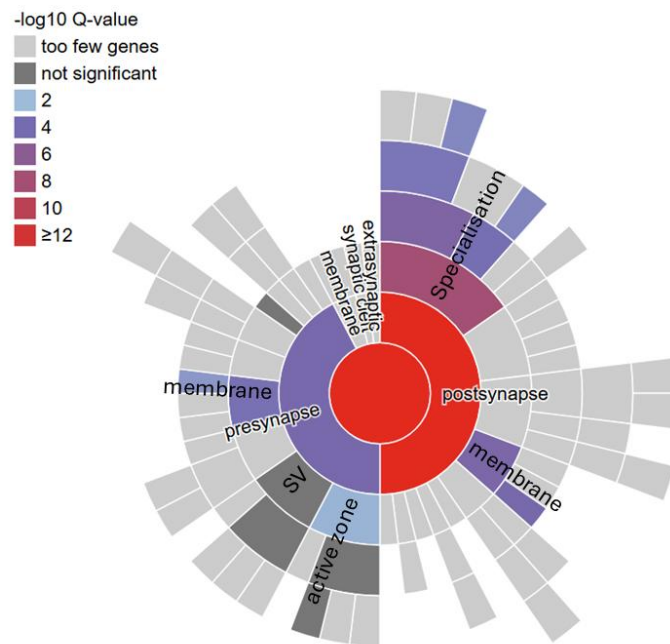


Figure 3. Sunburst plot of regions affected by DEGs -This figure illustrates the SynGO-based enrichment analysis of DEGs in schizophrenia, focusing on their synaptic localization. The analysis highlights significant enrichment of DEGs within specific synaptic compartments, reinforcing the hypothesis that synaptic dysfunction is a core feature of schizophrenia. (SV- synaptic vesicle)

7.4 MAGMA Analysis

The p-value ($P = 0.047$) of the DEGs indicates the statistical significance of the association between the overall gene set of DEGs and schizophrenia. A p-value of less than 0.05 suggests that the gene set is significantly enriched for genetic associations with schizophrenia. Among the genes analysed, 11 genes had a per-gene p-value below the Bonferroni correction threshold, indicating that these FOXP1-regulated genes are significantly associated with schizophrenia. These genes are considered high-confidence candidates and are likely to play important roles in the molecular mechanisms underlying the disorder. This finding highlights the relevance of the gene set in the context of schizophrenia and provides a strong foundation for further biological investigation.

These genes were *SPATS2L*, *SCAF1*, *NEGR1*, *ZNF408*, *HAPLN4*, *MAP1A*, *IMMP2L*, *CCDC62*, *SORCS3*, *ZNF408A*, *PDE4D*.

8. Individual Gene Analysis

To prioritize genes for further investigation, a LocusZoom analysis was performed to classify the most significantly associated genes based on their association signals in GWAS data. Genes were categorized into three groups: strong, moderate, and weak associations. Strongly associated genes included *SPATS2L*, *SCAF1*, *NEGR1*, *IMMP2L*, and *ZNF804A*. Moderately associated genes included *ZNF408*, *HAPLN4* and *MAP1A*. Genes with weak associations, such as *CCDC62*, *SORCS3*, and *PDE4D* were excluded from further analysis as their signals did not meet the threshold for continued investigation. The genes with strong and moderate associations were further examined to assess their potential biological relevance.

8.1 *SPATS2L*

SPATS2L exhibited a strong association signal, warranting further investigation (Fig.4). The most associated SNP, rs994280, was analysed using HaploReg to assess its functional significance and identify linked SNPs. This SNP was found to be located within an intronic region, suggesting a low likelihood of direct functional impact.

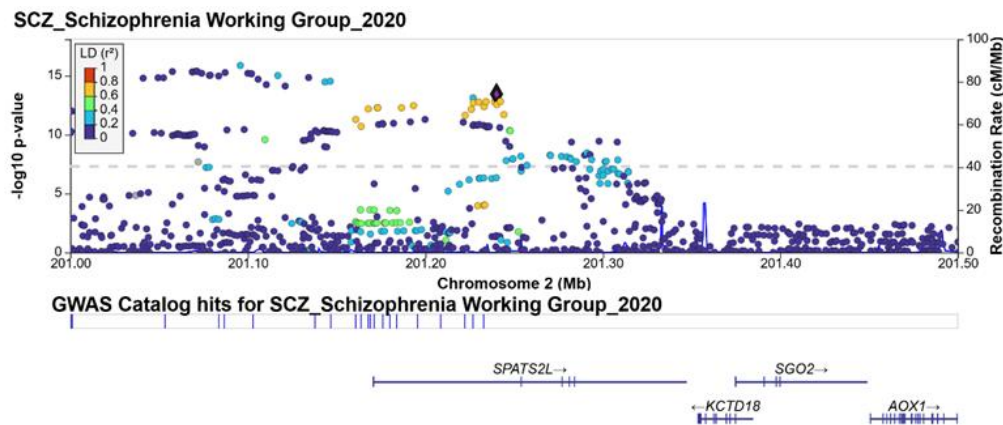


Figure 4. LocusZoom Plot for *SPATS2L*. This LocusZoom plot presents the genome-wide association study (GWAS) results for schizophrenia from the SCZ_Schizophrenia Working Group_2020, focusing on a locus on Chromosome 2. The lead SNP is located within the *SPATS2L* gene and exhibits a strong association signal, as indicated by its high $-\log_{10}$ p-value.

However, gene expression analysis using GTEx revealed that rs994280 influences *SPATS2L* expression in the brain, particularly within the cerebellum, indicating a potential role in neurological processes (Fig.5). The G allele of this SNP is associated with an increased risk of schizophrenia and decreases the expression of *SPATS2L* in cerebellum. This may indicate lower *SPATS2L* expression contributes to schizophrenia susceptibility.

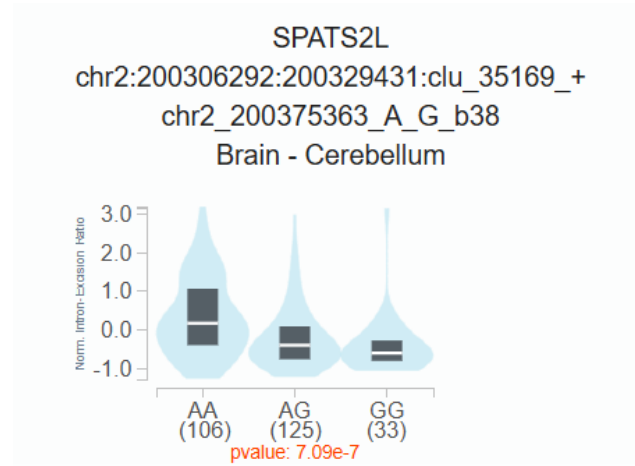


Figure 5. Violin plot for gene expression of SPATS2L SNP - This violin plot shows the effect of the chr2_200375363_A_G_b38 (rs994280) variant on SPATS2L gene's normalized intron-excision ratio in the Brain - Cerebellum. Genotypes AA, AG, and GG are compared, with AA showing the highest ratio and GG the lowest. The differences are statistically significant (p-value: 7.09e-7), suggesting a strong association between the variant and gene splicing.

8.2 ZNF804A

ZNF804A exhibited a strong association signal in the LocusZoom analysis (Fig.6). The most associated SNP, rs10196799, was found to be intronic with no direct functional effect. However, analysis of linked SNPs rs4667001 and rs1366842 both in low LD with rs10196799 revealed that they result in missense mutations. These variants were subsequently assessed with Ensembl VEP, which generated SIFT scores of 0.06 and 0.26 respectively. These values indicate that the mutations are tolerated rather than deleterious, suggesting that while they may influence protein function their impact on disease pathology is likely to be minimal.

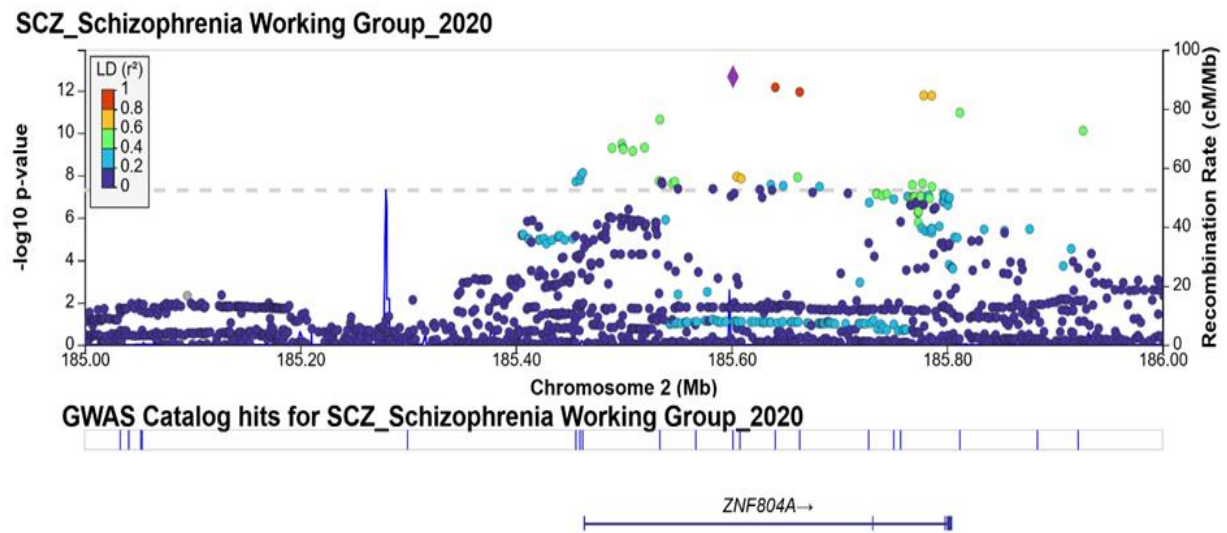


Figure 6. LocusZoom Plot for ZNF804A- This LocusZoom plot presents the genome-wide association study (GWAS) results for schizophrenia from the SCZ_Schizophrenia Working Group_2020, focusing on a locus on Chromosome 2. The lead SNP is located within the *ZNF804A* gene and exhibits a strong association signal, as indicated by its high $-\log_{10}$ p-value.

8.3 NEGR1

NEGR1 was assigned a strong classification, given its strong association signal in the LocusZoom plot (Fig.7). The most associated SNP, rs11209828, was identified as intronic in HaploReg, indicating no clear functional consequence. Gene expression analysis via GTEx also showed no significant regulatory effect. Despite the strong association signal, there is a lack of functional evidence to explain the effect of the variant on *NEGR1*.

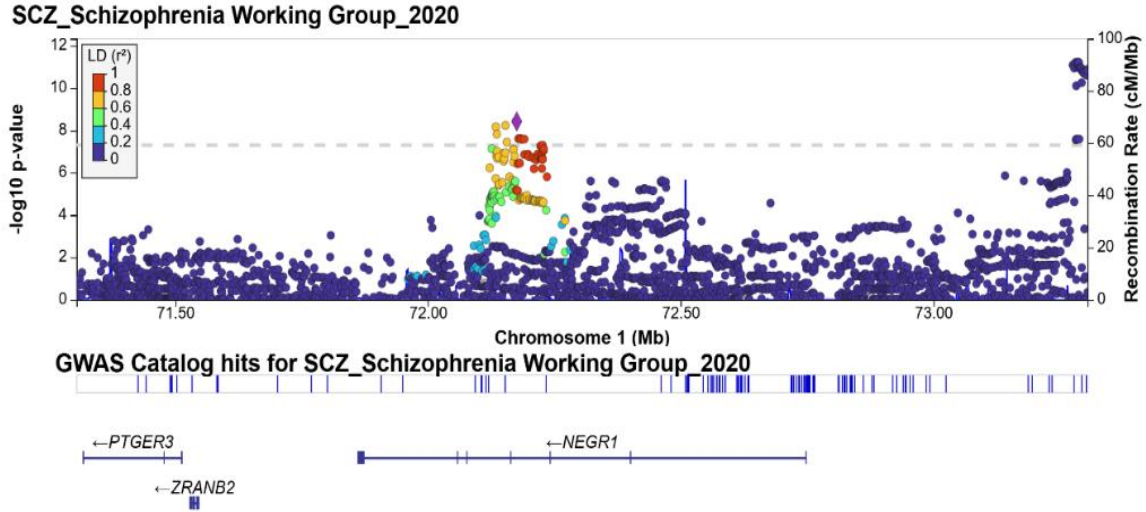


Figure 7. Locuszoom plot of NEGR1- This plot shows the association between genetic variants near NEGR1 and schizophrenia (SCZ) using data from the SCZ_Schizophrenia Working Group_2020. A significant association is observed near NEGR1, suggesting its potential role in schizophrenia

8.4 HAPLN4

HAPLN4 was classified as moderate due to its presence in a region of interest but without direct evidence of association with schizophrenia. The most associated SNP, rs56135441, was intronic, suggesting no obvious functional consequences. However, GTEx analysis showed that this SNP influences gene expression in three distinct brain regions indicating that it may contribute to schizophrenia risk through regulatory mechanisms affecting neural function (Fig.8) The T allele of this SNP is associated with schizophrenia risk and decreases the expression of *HAPLN4* in the cerebellum, frontal cortex and cerebellar hemisphere. This may indicate lower *HAPLN4* expression contributes to schizophrenia susceptibility.

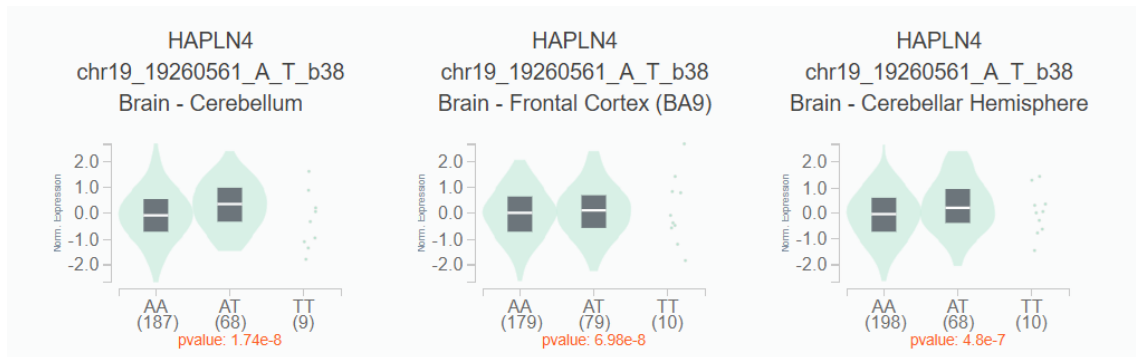


Figure 8. Violin plot of HAPLN4 gene expression. This figure shows HAPLN4 gene expression in three brain regions: Cerebellum, Frontal Cortex (BA9), and Cerebellar Hemisphere, influenced by the chr19_19260561_A_T_b38 (rs56135441) variant. Significant associations are seen in all regions with varying expression levels across genotypes. The variant significantly affects HAPLN4 expression in these brain regions.

8.5 MAP1A

MAP1A was categorised as moderate, as it was in a region of high schizophrenia risk association, though direct evidence linking it to the disease was lacking. The lead SNP, rs1869258, was found to be intronic, suggesting no direct protein-coding effects. However, GTEx analysis revealed that this SNP affects gene expression in the basal ganglia and cerebellar hemisphere (Fig.9), both of which have been implicated in schizophrenia pathophysiology. These findings suggest a potential, albeit indirect, role for *MAP1A* in the disorder. The A allele is associated with schizophrenia risk and decrease the expression of *MAP1A* in the putamen and cerebellar hemisphere. This would suggest that reduced *MAP1A* expression contributes to schizophrenia risk.

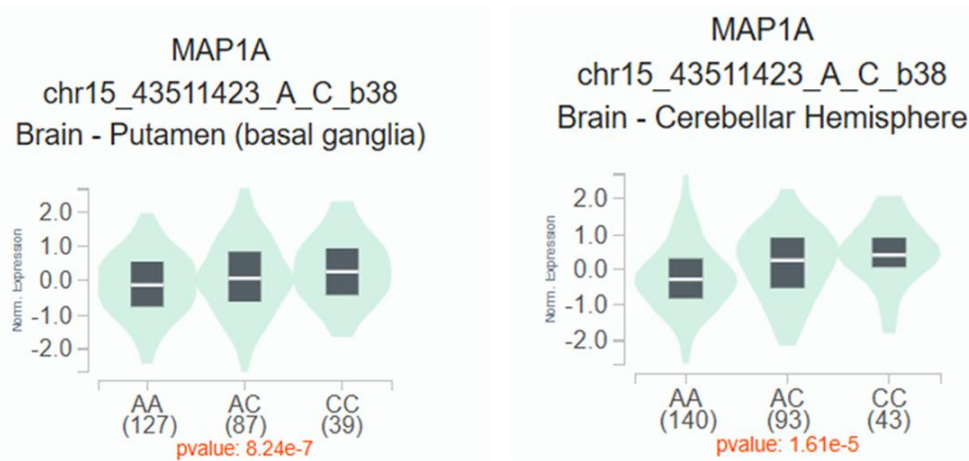


Figure 9 Violin plot of *MAP1A* gene expression. The *MAP1A* variant chr15_43511423_A_C_b38 (rs1869258) shows significant eQTL associations in Putamen ($p=8.24e-7$) and Cerebellar Hemisphere ($p=1.61e-5$). Boxplots display expression levels by genotype (AA/AC/CC), with sample sizes labeled. Dashed lines indicate normalized expression trends.

8.6 SCAF1, ZNF408 AND IMMPL2

SCAF1 and *ZNF408* were categorised as moderate-priority genes due to unclear association signals in locus zoom analysis and limited evidence of brain-related effects. The most associated SNPs, rs7047 (*SCAF1*) and rs79175663 (*ZNF408*), were found in non-coding regions, with gene expression analyses showing no significant impact on the brain. *IMMP2L* was considered a high-priority gene due to its strong schizophrenia association; however, its top SNP rs38752 was intronic and GTEx analysis showed no significant brain expression effects.

9. Discussion

This study was an extension of previous *FOXP1* cKO mouse model research, which initially focused on neurodevelopmental disorders such as autism spectrum disorder and intellectual disability. The original murine studies demonstrated *FOXP1*'s crucial role in neuronal development, particularly in synaptic function and hippocampal connectivity. This investigation specifically addresses the unexplored potential of these findings in the context of schizophrenia.

By leveraging the transcriptomic data from the original cKO mouse models, advanced bioinformatics tools were applied to systematically explore how *FOXP1* regulated gene networks intersect with schizophrenia risk loci. This approach transformed preliminary mouse model observations into a targeted investigation of schizophrenia's molecular mechanisms.

9.1 Candidate genes – insights

Analysis of individual genes highlights their potential roles in schizophrenia through their involvement in synaptic structure and function. *MAP1A*, associated with both the postsynaptic density and the presynaptic active zone, plays a crucial role in maintaining synaptic integrity. *HAPLN4A*, linked to synapses, may influence synaptic organization, and its dysregulation could impact neural connectivity, a key feature implicated in the disorder. *NEGR1*, localized to the postsynaptic density is associated with synaptic signaling and plasticity, processes that are often impaired in schizophrenia. These findings were identified through SYNGO analysis, emphasizing the relevance of these genes in synaptic architecture and function, which may contribute to the neurodevelopmental and connectivity deficits underlying schizophrenia.

NEGR1 showed strong statistical association despite limited functional evidence. This cell adhesion molecule mediates neurite outgrowth and synapse formation during neurodevelopment, aligning with the neurodevelopmental hypothesis of schizophrenia. Previous studies have linked *NEGR1* variants to major depression suggesting its involvement in psychiatric pathways [20]. The co-occurrence of metabolic disturbances in schizophrenia is also noteworthy as *NEGR1* has been implicated in metabolic regulation [21].

ZNF804A has been consistently implicated in schizophrenia across multiple GWAS. Recent comprehensive reviews have examined its clinical associations and biological mechanisms [22]. As a transcription factor, *ZNF804A* modulates genes involved in cell adhesion and neurite outgrowth. Neuroimaging studies show *ZNF804A* variants affect functional connectivity between brain regions crucial for working memory and executive function aligning with the disconnection

hypothesis of schizophrenia [22]. The identification of tolerated missense mutations suggests subtle alterations in *ZNF804A* function may contribute to disease risk.

Despite its classification as moderate, *HAPLN4* showed significant expression effects in multiple brain regions, including the cerebellum, frontal cortex, and cerebellar hemisphere. *HAPLN4* encodes a protein involved in the formation and maintenance of perineuronal nets which are specialized extracellular matrix structures that stabilize synapses and regulate neural plasticity [23]. Perineuronal nets have been implicated in schizophrenia pathophysiology, with postmortem studies reporting reduced density in the prefrontal cortex of schizophrenia patients [24]. The finding of altered *HAPLN4* expression across multiple brain regions suggests potential dysregulation of perineuronal nets, which could contribute to the synaptic abnormalities and altered neural plasticity observed in schizophrenia.

MAP1A, showing significant expression effects on the hippocampus, represents a compelling candidate. This microtubule-associated protein is critical for neuronal cytoskeletal stability, dendritic development, and synaptic plasticity [25]. The hippocampus, where expression changes were observed, is central to learning and memory domains frequently disrupted in schizophrenia. *MAP1A* has been implicated in neurodevelopmental disorders due to its role in maintaining neuronal structure and function. It interacts with synaptic proteins linked to schizophrenia pathophysiology, including those involved in glutamatergic signaling and postsynaptic density organization [26].

9.2 Limitations and Challenges

This study faces several key challenges in translating findings from research into practical insights. A major limitation is the gap between mouse models and human neurological complexity. While mouse models offer valuable insights into genetic mechanisms, they cannot fully replicate the intricate neural circuits and cognitive functions of the human brain. Differences in brain structure, neuronal connectivity, and gene expression between species require careful interpretation of results. Additionally, the bioinformatic predictions in this study, though computationally robust, need experimental validation. Schizophrenia involves complex interactions between multiple neural networks, and current methods struggle to fully map these genetic interactions. Each brain region has distinct gene expression patterns, making it difficult to form a complete picture of the disorder. These challenges highlight the importance of combining computational analysis, animal models, and human genetic studies to better understand the genetic basis of schizophrenia.

10. Conclusion

This comprehensive study represents a significant advancement in understanding the genetic underpinnings of schizophrenia by bridging mouse model research with human genetic analysis. By integrating RNA sequencing data, GWAS and advanced bioinformatics tools, insights into the molecular mechanisms potentially contributing to schizophrenia's complex pathophysiology could be uncovered. This investigation highlighted several key candidate genes, most notably *FOXP1*, which emerged as a compelling target for further research. The transcription factor's role in neurodevelopment, particularly in hippocampal gene regulation, underscores its potential significance in schizophrenia. Other genes like *NEGR1*, *ZNF804A*, and *HAPLN4* also demonstrated intriguing associations, suggesting diverse molecular pathways involved in the disorder. The SynGO analysis revealed a profound enrichment of synaptic components and processes among DEGs, reinforcing the neurobiological hypothesis that synaptic dysfunction is central to schizophrenia. This finding provides a crucial molecular perspective on the disorder's underlying mechanisms. Ultimately, this research enhances understanding of schizophrenia's genetic architecture, highlighting the need for multi-modal approaches that combine computational analysis, animal models, and human genetic studies to uncover the disorder's molecular foundations.

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Appendices

Github- <https://github.com/bogh1/Research-Project/blob/main/Code>

Supplementary Table 1. Full List Of DEGs

Ensembl gene identifier	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	mouse_gene_symbol	human_gene_symbol
ENSMUSG00000078612	69.26617	-3.12134	0.29596	-10.5465	5.27E-26	8.43E-22	Fyb2	FYB2
ENSMUSG00000026062	272.4993	-1.88502	0.190957	-9.87144	5.54E-23	4.43E-19	Slc9a2	SLC9A2
ENSMUSG00000025813	1655.676	-1.17034	0.120272	-9.73072	2.23E-22	1.19E-18	Homer2	HOMER2
ENSMUSG00000015568	1019.814	-1.18394	0.122643	-9.65359	4.75E-22	1.90E-18	Lpl	LPL
ENSMUSG00000026065	194.4484	-2.5984	0.272212	-9.54549	1.35E-21	4.33E-18	Slc9a4	SLC9A4
ENSMUSG00000002944	83.06914	2.765459	0.292218	9.463677	2.97E-21	7.92E-18	Cd36	CD36
ENSMUSG00000063531	1731.196	-1.10801	0.125991	-8.79437	1.44E-18	3.29E-15	Sema3e	SEMA3E
ENSMUSG000000103334	61.45853	4.020763	0.461949	8.703902	3.21E-18	6.41E-15	Gm29771	
ENSMUSG00000036095	6724.135	-0.64375	0.074095	-8.68826	3.68E-18	6.54E-15	Dgkb	DGKB
ENSMUSG000000027971	467.8414	-1.54712	0.191081	-8.09668	5.65E-16	9.03E-13	Ndst4	NDST4
ENSMUSG00000046607	815.9581	1.195251	0.152305	7.847765	4.24E-15	6.16E-12	Hrk	HRK
ENSMUSG00000026616	46.73007	5.401057	0.694182	7.780461	7.23E-15	8.89E-12	Cr2	CR2
ENSMUSG00000037578	49.15359	3.559799	0.457329	7.783883	7.03E-15	8.89E-12	Pkd2l1	PKD2L1
ENSMUSG00000003657	394.2693	2.492583	0.323523	7.704508	1.31E-14	1.50E-11	Calb2	CALB2
ENSMUSG00000021676	2451.487	-1.05295	0.140388	-7.50027	6.37E-14	6.79E-11	Iqgap2	IQGAP2
ENSMUSG00000070866	998.1633	-0.55546	0.077939	-7.12685	1.03E-12	1.03E-09	Zfp804a	ZNF804A
ENSMUSG000000040452	371.5531	-0.98379	0.140683	-6.99291	2.69E-12	2.53E-09	Cdh12	CDH12
ENSMUSG00000041225	1504.754	-0.82877	0.120011	-6.90576	4.99E-12	4.43E-09	Arhgap12	ARHGAP12
ENSMUSG00000044847	465.9332	-0.89135	0.130803	-6.8144	9.47E-12	7.96E-09	Lsm11	LSM11
ENSMUSG00000028931	2886.075	-0.93811	0.138302	-6.78307	1.18E-11	9.40E-09	Kcnab2	KCNAB2
ENSMUSG00000090125	329.8184	-1.45099	0.215941	-6.71939	1.82E-11	1.39E-08	Pou3f1	POU3F1
ENSMUSG00000036452	1541.649	-0.62322	0.093027	-6.69938	2.09E-11	1.52E-08	Arhgap26	ARHGAP26
ENSMUSG00000028033	2153.563	-0.55766	0.084079	-6.63257	3.30E-11	2.29E-08	Kcnq5	KCNQ5
ENSMUSG00000022594	1111.174	-0.53789	0.082281	-6.53717	6.27E-11	4.18E-08	Lynx1	LYNX1
ENSMUSG00000032925	300.9498	1.206854	0.186522	6.470302	9.78E-11	6.25E-08	Itgbl1	ITGBL1
ENSMUSG00000074934	123.5677	-1.17622	0.184619	-6.37106	1.88E-10	1.15E-07	Grem1	GREM1
ENSMUSG00000009075	526.0406	-0.97283	0.153235	-6.34862	2.17E-10	1.29E-07	Cabp7	CABP7
ENSMUSG00000045053	170.4311	1.01897	0.163028	6.250284	4.10E-10	2.34E-07	Kcng3	KCNG3
ENSMUSG00000021313	5088.492	-0.58611	0.098923	-5.92493	3.12E-09	1.72E-06	Ryr2	RYR2
ENSMUSG00000025885	1141.987	-0.89271	0.150946	-5.91412	3.34E-09	1.78E-06	Myo5b	MYO5B
ENSMUSG00000037138	831.4486	-0.58062	0.100392	-5.78354	7.31E-09	3.77E-06	Aff3	AFF3
ENSMUSG00000059742	1996.21	-0.85735	0.148776	-5.76269	8.28E-09	4.14E-06	Kcnh7	KCNH7
ENSMUSG00000047415	112.1884	-0.95693	0.166547	-5.74569	9.15E-09	4.44E-06	Gpr68	GPR68
ENSMUSG00000030782	86.02707	1.099667	0.193036	5.696691	1.22E-08	5.74E-06	Tgfb1i1	TGFB1I1

ENSMUSG00000010803	5906.709	-0.41609	0.073437	-5.6659	1.46E-08	6.68E-06	Gabra1	GABRA1
ENSMUSG00000032036	453.6764	0.702636	0.124696	5.634772	1.75E-08	7.74E-06	Kirrel3	KIRREL3
ENSMUSG00000063434	690.7563	-0.60517	0.107469	-5.6311	1.79E-08	7.74E-06	Sorcs3	SORCS3
ENSMUSG00000033717	168.8804	1.278392	0.227234	5.625886	1.85E-08	7.76E-06	Adra2a	ADRA2A
ENSMUSG000000100166	18.05628	-4.68706	0.840497	-5.57653	2.45E-08	1.01E-05	B230110G15Rik	
ENSMUSG00000030067	1986.788	-1.08073	0.195075	-5.54008	3.02E-08	1.18E-05	Foxp1	FOXP1
ENSMUSG00000075590	1410.448	0.391109	0.070599	5.539884	3.03E-08	1.18E-05	Nrbp2	NRBP2
ENSMUSG00000030352	201.8327	0.895383	0.16176	5.535246	3.11E-08	1.18E-05	Tspan9	TSPAN9
ENSMUSG000000003746	1728.719	-0.64203	0.116086	-5.53062	3.19E-08	1.19E-05	Man1a	MAN1A1
ENSMUSG00000063600	70.44817	-1.76181	0.322579	-5.46164	4.72E-08	1.64E-05	Egfm1	
ENSMUSG00000005087	132.5132	1.386676	0.253857	5.462422	4.70E-08	1.64E-05	Cd44	CD44
ENSMUSG00000032502	18.20503	-2.55862	0.467824	-5.46919	4.52E-08	1.64E-05	Stac	STAC
ENSMUSG00000024769	62.38385	3.191984	0.587169	5.436229	5.44E-08	1.85E-05	Cdc42bpg	CDC42BPG
ENSMUSG00000053279	625.1455	-1.02582	0.189444	-5.4149	6.13E-08	2.01E-05	Aldh1a1	ALDH1A1
ENSMUSG00000042514	67.13012	1.603003	0.296085	5.413996	6.16E-08	2.01E-05	Klhl14	KLHL14
ENSMUSG000000040105	837.3771	-0.49286	0.091616	-5.37959	7.47E-08	2.39E-05	Plpp6	PLPP6
ENSMUSG00000030499	56.81294	-3.55187	0.664424	-5.34579	9.00E-08	2.78E-05	Kctd15	KCTD15
ENSMUSG00000028833	3265.283	-0.49669	0.092927	-5.34494	9.04E-08	2.78E-05	Ncdn	NCDN
ENSMUSG00000046159	346.8993	-1.08293	0.204462	-5.29648	1.18E-07	3.56E-05	Chrm3	CHRM3
ENSMUSG00000040536	4701.048	0.968246	0.184748	5.240912	1.60E-07	4.73E-05	Necab1	NECAB1
ENSMUSG00000028020	3000.797	-0.33104	0.063568	-5.20759	1.91E-07	5.56E-05	Glrb	GLRB
ENSMUSG00000033491	138.7265	-0.97905	0.188244	-5.20098	1.98E-07	5.66E-05	Prss35	PRSS35
ENSMUSG000000047139	504.848	1.02681	0.20005	5.132758	2.86E-07	8.01E-05	Cd24a	CD24
ENSMUSG00000090272	318.9767	0.67109	0.131187	5.115539	3.13E-07	8.62E-05	Mndal	MNDA
ENSMUSG00000066113	166.8013	1.591434	0.311571	5.107782	3.26E-07	8.65E-05	Adamts1	ADAMTSL1
ENSMUSG00000032564	2106.093	-0.5514	0.108004	-5.10533	3.30E-07	8.65E-05	Cpne4	CPNE4
ENSMUSG00000051359	5152.526	-0.5346	0.104587	-5.11156	3.20E-07	8.65E-05	Ncald	NCALD
ENSMUSG00000073418	214.0501	0.932224	0.182878	5.097509	3.44E-07	8.80E-05	C4b	C4A
ENSMUSG00000032262	1547.427	-0.45106	0.088511	-5.09616	3.47E-07	8.80E-05	Elovl4	ELOVL4
ENSMUSG00000027254	14426.63	-0.28835	0.056906	-5.06716	4.04E-07	0.000101	Map1a	MAP1A
ENSMUSG00000028028	140.9977	-1.22732	0.243782	-5.03448	4.79E-07	0.000118	Alpk1	ALPK1
ENSMUSG00000032479	10357.97	-0.34463	0.068553	-5.0272	4.98E-07	0.000121	Map4	MAP4
ENSMUSG00000060882	2564.365	-0.56882	0.113304	-5.02034	5.16E-07	0.000121	Kcnd2	KCND2
ENSMUSG00000030605	376.6692	0.68271	0.135963	5.021282	5.13E-07	0.000121	Mfge8	MFGE8
ENSMUSG00000038602	1693.416	0.557987	0.112081	4.978423	6.41E-07	0.000149	Slc35f1	SLC35F1
ENSMUSG00000074971	86.261	1.583635	0.321227	4.92996	8.22E-07	0.000188	Fibin	FIBIN
ENSMUSG000000020380	589.4694	0.610375	0.124289	4.910923	9.06E-07	0.000202	Rad50	RAD50
ENSMUSG00000049907	192.5413	-0.76696	0.156189	-4.91046	9.09E-07	0.000202	Rasl11b	RASL11B
ENSMUSG000000041439	1446.883	-0.32868	0.067164	-4.89362	9.90E-07	0.000217	Mfsd6	MFSD6
ENSMUSG00000029189	437.8697	-0.54261	0.111162	-4.88123	1.05E-06	0.000228	Set1l3	SEL1L3
ENSMUSG00000003948	6272.343	-0.50676	0.103936	-4.87566	1.08E-06	0.000231	Mmd	MMD
ENSMUSG00000033350	2508.31	-0.43288	0.089107	-4.85802	1.19E-06	0.000248	Chst2	CHST2

ENSMUSG00000054074	219.2968	-0.81889	0.168603	-4.85692	1.19E-06	0.000248	Skida1	SKIDA1
ENSMUSG00000023169	2730.007	-0.31777	0.065579	-4.84559	1.26E-06	0.000256	Slc38a1	SLC38A1
ENSMUSG00000049577	76.30316	-1.50181	0.309949	-4.84536	1.26E-06	0.000256	Zfpn1	ZFPM1
ENSMUSG00000020439	59.77425	1.98308	0.410021	4.836529	1.32E-06	0.000264	Smtn	SMTN
ENSMUSG00000027350	6288.669	-0.53947	0.111736	-4.82809	1.38E-06	0.000272	Chgb	CHGB
ENSMUSG00000028266	2527.702	0.45119	0.093822	4.809017	1.52E-06	0.000292	Lmo4	LMO4
ENSMUSG0000002997	1229.714	0.490362	0.101961	4.809301	1.51E-06	0.000292	Prkar2b	PRKAR2B
ENSMUSG00000071753	11227.47	-0.51205	0.107317	-4.7714	1.83E-06	0.000345	Cdr1os	
ENSMUSG00000027931	56.28995	1.650478	0.345946	4.770918	1.83E-06	0.000345	Npr1	NPR1
ENSMUSG00000079110	78.58982	-2.06071	0.43303	-4.75882	1.95E-06	0.000362	Capn3	CAPN3
ENSMUSG00000042298	1299.295	-0.38709	0.082211	-4.70843	2.50E-06	0.000459	Ttc19	TTC19
ENSMUSG00000028601	59.37845	1.281625	0.272655	4.700544	2.59E-06	0.000471	Echdc2	ECHDC2
ENSMUSG00000028785	5901.48	-0.67977	0.144784	-4.69506	2.67E-06	0.000476	Hpca	HPCA
ENSMUSG00000040037	6029.821	0.639471	0.13623	4.694048	2.68E-06	0.000476	Negr1	NEGR1
ENSMUSG00000042156	1644.213	-0.41676	0.088898	-4.68808	2.76E-06	0.000484	Dzip1	DZIP1
ENSMUSG00000021701	1883.829	-0.56096	0.119803	-4.68237	2.84E-06	0.000493	Plk2	PLK2
ENSMUSG00000033389	720.9649	-0.40916	0.087721	-4.66432	3.10E-06	0.000532	Arhgap44	ARHGAP44
ENSMUSG00000029174	401.649	-0.70773	0.153277	-4.61731	3.89E-06	0.000661	Tbc1d1	TBC1D1
ENSMUSG000000104586	87.33302	-2.06835	0.449219	-4.60434	4.14E-06	0.000696	4921539H07Rik	
ENSMUSG00000075302	20.036	2.154639	0.468262	4.601357	4.20E-06	0.000699	Erich2	ERICH2
ENSMUSG00000022519	103.7565	-1.14163	0.250931	-4.54959	5.37E-06	0.000886	Srl	SRL
ENSMUSG00000061576	3534.587	-0.34581	0.076121	-4.54289	5.55E-06	0.000905	Dpp6	DPP6
ENSMUSG00000050587	2416.756	-0.46817	0.103258	-4.53393	5.79E-06	0.000921	Lrrc4c	LRRC4C
ENSMUSG00000059857	440.5385	-0.88167	0.19438	-4.53583	5.74E-06	0.000921	Ntng1	NTNG1
ENSMUSG00000048070	21.66142	-2.08523	0.460028	-4.53282	5.82E-06	0.000921	Pirt	PIRT
ENSMUSG00000043668	233.4521	0.92384	0.204567	4.516063	6.30E-06	0.000987	Tox3	TOX3
ENSMUSG00000021803	109.8975	0.821453	0.182037	4.512571	6.40E-06	0.000994	Cdhr1	CDHR1
ENSMUSG00000025089	713.6875	0.427656	0.09521	4.491726	7.06E-06	0.001086	Gfra1	GFRA1
ENSMUSG00000049892	205.9401	-1.27454	0.284158	-4.48534	7.28E-06	0.001108	Rasd1	RASD1
ENSMUSG00000032290	432.3599	-0.53026	0.118291	-4.48267	7.37E-06	0.001112	Ptpn9	PTPN9
ENSMUSG00000040183	625.7893	0.543761	0.121997	4.457176	8.30E-06	0.001241	Ankrd6	ANKRD6
ENSMUSG00000000631	1010.472	-0.50153	0.112658	-4.45177	8.52E-06	0.001261	Myo18a	MYO18A
ENSMUSG00000019785	549.9345	-0.69369	0.155903	-4.4495	8.61E-06	0.001262	Clvs2	CLVS2
ENSMUSG00000027577	156.0148	0.986232	0.222379	4.434912	9.21E-06	0.001327	Chrna4	CHRNA4
ENSMUSG00000026604	304.1968	0.812874	0.183295	4.434776	9.22E-06	0.001327	Ptpn14	PTPN14
ENSMUSG00000039497	114.4207	-2.06683	0.468447	-4.41208	1.02E-05	0.001448	Dse	DSE
ENSMUSG00000021027	3459.705	-0.41305	0.093587	-4.4135	1.02E-05	0.001448	Ralgapa1	RALGAPA1
ENSMUSG00000039985	143.9163	-0.90959	0.206338	-4.40824	1.04E-05	0.001461	Sinhcaf	SINHCAF
ENSMUSG00000046442	10511.05	-0.57701	0.131038	-4.40335	1.07E-05	0.001482	Ppm1e	PPM1E
ENSMUSG00000026354	206.1489	-1.24203	0.282502	-4.39654	1.10E-05	0.001509	Lct	LCT
ENSMUSG00000031558	1267.36	-0.58618	0.133354	-4.39563	1.10E-05	0.001509	Slit2	SLIT2
ENSMUSG00000030102	5085.016	-0.62289	0.141926	-4.38881	1.14E-05	0.001544	Itpr1	ITPR1

ENSMUSG00000002799	155.6419	1.181589	0.270215	4.372775	1.23E-05	0.001648	Jag2	JAG2
ENSMUSG000000032012	650.0833	0.923788	0.211505	4.367678	1.26E-05	0.001673	Nectin1	NECTIN1
ENSMUSG000000057897	8686.676	-0.3868	0.089082	-4.34206	1.41E-05	0.001865	Camk2b	CAMK2B
ENSMUSG000000034765	116.5953	-1.04097	0.239982	-4.33769	1.44E-05	0.001887	Dusp5	DUSP5
ENSMUSG000000032487	552.6369	-0.76813	0.177439	-4.32899	1.50E-05	0.001947	Ptgs2	PTGS2
ENSMUSG000000049690	148.0114	0.716869	0.165995	4.318607	1.57E-05	0.00201	Nckap5	NCKAP5
ENSMUSG000000017843	3037.965	-0.31712	0.073436	-4.31837	1.57E-05	0.00201	Ppp2r5c	PPP2R5C
ENSMUSG000000097789	1807.841	-0.95102	0.22043	-4.31438	1.60E-05	0.002031	Gm2115	CTXND1
ENSMUSG000000071317	195.5258	0.857327	0.200254	4.281187	1.86E-05	0.00234	Bves	BVES
ENSMUSG000000036862	108.3089	-1.00869	0.235994	-4.27423	1.92E-05	0.002396	Dchs1	DCHS1
ENSMUSG000000032373	1197.829	0.571381	0.134229	4.256766	2.07E-05	0.002551	Car12	CA12
ENSMUSG000000001334	676.1755	-0.51198	0.120263	-4.25714	2.07E-05	0.002551	Fndc5	FNDC5
ENSMUSG000000019880	242.6668	1.014082	0.23913	4.240717	2.23E-05	0.002719	Rspo3	RSPO3
ENSMUSG000000029178	1169.305	-0.68463	0.161586	-4.23696	2.27E-05	0.002744	Klf3	KLF3
ENSMUSG000000023030	326.1056	0.750512	0.178432	4.206147	2.60E-05	0.003122	Slc11a2	SLC11A2
ENSMUSG000000047259	52.75071	-1.33766	0.318561	-4.19907	2.68E-05	0.003174	Mc4r	MC4R
ENSMUSG000000059991	109.8147	-0.6935	0.165094	-4.20062	2.66E-05	0.003174	Nptx2	NPTX2
ENSMUSG000000024479	3497.247	-0.32515	0.077582	-4.19108	2.78E-05	0.003264	Mal2	MAL2
ENSMUSG000000028351	2403.881	-0.43479	0.103953	-4.18257	2.88E-05	0.003339	Brinp1	BRINP1
ENSMUSG000000029432	845.014	1.206655	0.288473	4.182902	2.88E-05	0.003339	Nipsnap2	NIPSNAP2
ENSMUSG000000074521	75.06573	-0.94306	0.225755	-4.17736	2.95E-05	0.003392	Gm14327	
ENSMUSG000000050321	7047.903	-0.56622	0.135681	-4.17316	3.00E-05	0.003415	Neto1	NETO1
ENSMUSG000000033933	495.2554	0.503642	0.120704	4.172556	3.01E-05	0.003415	Vhl	VHL
ENSMUSG000000044647	2377.117	-0.33766	0.080966	-4.17039	3.04E-05	0.003423	Csrnp3	CSRNP3
ENSMUSG000000091345	67.46048	1.636568	0.393569	4.158279	3.21E-05	0.00356	Col6a5	COL6A5
ENSMUSG000000045095	1785.088	0.34952	0.084034	4.159278	3.19E-05	0.00356	Magi1	MAGI1
ENSMUSG000000053414	250.493	-0.67193	0.162221	-4.14208	3.44E-05	0.003795	Hunk	HUNK
ENSMUSG000000036242	575.6304	-0.44054	0.106405	-4.14022	3.47E-05	0.003799	Armh4	ARMH4
ENSMUSG000000022382	146.1327	0.659373	0.160159	4.11699	3.84E-05	0.004175	Wnt7b	WNT7B
ENSMUSG000000025555	629.4525	0.576838	0.141405	4.07932	4.52E-05	0.004879	Farp1	FARP1
ENSMUSG000000086040	6111.869	-0.78264	0.192004	-4.07618	4.58E-05	0.004912	Wipf3	WIPF3
ENSMUSG000000107677	18.257	1.763986	0.433012	4.073755	4.63E-05	0.004931	Olfir316	OR2AK2
ENSMUSG000000054263	1625.575	-0.45405	0.111759	-4.0628	4.85E-05	0.005134	Lifr	LIFR
ENSMUSG000000058656	1823.998	-0.36522	0.090054	-4.05559	5.00E-05	0.005258	Samd12	SAMD12
ENSMUSG000000021217	208.9879	0.818298	0.201844	4.054114	5.03E-05	0.005258	Tshz3	TSHZ3
ENSMUSG000000027966	318.0861	-0.8097	0.199974	-4.04904	5.14E-05	0.005339	Col11a1	COL11A1
ENSMUSG000000097881	36.47057	1.230139	0.304276	4.042839	5.28E-05	0.005412	Celrr	
ENSMUSG000000026721	6940.958	-0.52232	0.129151	-4.04424	5.25E-05	0.005412	Rabgap1l	RABGAP1L
ENSMUSG000000027495	329.9564	-0.54451	0.134736	-4.04129	5.32E-05	0.005413	Fam210b	FAM210B
ENSMUSG000000062908	453.8056	0.500234	0.124135	4.029756	5.58E-05	0.00565	Acadm	ACADM
ENSMUSG000000026322	203.1641	-0.70005	0.174057	-4.02195	5.77E-05	0.005767	Htr4	HTR4
ENSMUSG000000056258	6815.002	-0.47907	0.119109	-4.0221	5.77E-05	0.005767	Kcnq3	KCNQ3

ENSMUSG00000055078	3158.442	-0.51276	0.127613	-4.01806	5.87E-05	0.005827	Gabra5	GABRA5
ENSMUSG00000020953	163.3329	0.959351	0.240017	3.997004	6.41E-05	0.006331	Coch	COCH
ENSMUSG00000000223	854.4836	-0.65572	0.164272	-3.99168	6.56E-05	0.006408	Drp2	DRP2
ENSMUSG00000042249	1972.141	0.459269	0.115069	3.991229	6.57E-05	0.006408	Grk3	GRK3
ENSMUSG00000021866	223.5948	-0.55762	0.139857	-3.98711	6.69E-05	0.006444	Anxa11	ANXA11
ENSMUSG00000026839	285.5494	1.135494	0.284799	3.987002	6.69E-05	0.006444	Upp2	UPP2
ENSMUSG00000006800	940.5956	-0.37292	0.093708	-3.97963	6.90E-05	0.006608	Sulf2	SULF2
ENSMUSG000000111971	66.92849	1.651536	0.415169	3.97799	6.95E-05	0.006614	Gm48678	
ENSMUSG000000105703	7950.047	-0.39199	0.098716	-3.97088	7.16E-05	0.006734	Gm43305	
ENSMUSG000000066406	1626.961	-0.70786	0.178252	-3.97112	7.15E-05	0.006734	Akap13	AKAP13
ENSMUSG00000042810	444.9957	0.702696	0.177079	3.968272	7.24E-05	0.006768	Krba1	KRBA1
ENSMUSG00000022797	1869.682	-0.35282	0.089036	-3.96266	7.41E-05	0.006889	Tfrc	TFRC
ENSMUSG00000004110	12952.03	-0.38105	0.096491	-3.94911	7.84E-05	0.007249	Cacna1e	CACNA1E
ENSMUSG00000059244	2446.553	-0.46692	0.118623	-3.93618	8.28E-05	0.007607	Gm10062	
ENSMUSG00000046532	1002.89	-0.59692	0.151762	-3.93329	8.38E-05	0.007634	Ar	AR
ENSMUSG00000026072	238.7473	-0.86289	0.219421	-3.93258	8.40E-05	0.007634	Il1r1	IL1R1
ENSMUSG00000085039	27.90027	3.189341	0.811358	3.930866	8.46E-05	0.007645	Gm15927	
ENSMUSG00000026786	167.2434	0.67551	0.172448	3.917176	8.96E-05	0.008047	Apbb1ip	APBB1IP
ENSMUSG00000002808	726.6228	-0.48828	0.124857	-3.91067	9.20E-05	0.008196	Epdr1	EPDR1
ENSMUSG00000034810	188.7544	0.880742	0.22525	3.910055	9.23E-05	0.008196	Scn7a	SCN7A
ENSMUSG00000046709	4777.052	-0.45718	0.117179	-3.90155	9.56E-05	0.008412	Mapk10	MAPK10
ENSMUSG00000025889	2224.649	0.389787	0.099918	3.901067	9.58E-05	0.008412	Snca	SNCA
ENSMUSG000000107272	48.2663	-1.23242	0.316533	-3.89349	9.88E-05	0.008632	Gm34583	
ENSMUSG00000027419	2557.526	-0.50714	0.130445	-3.8878	0.000101	0.008789	Pcsk2	PCSK2
ENSMUSG00000072893	92.52088	0.694903	0.179317	3.875277	0.000107	0.009204	4933439C10Rik	
ENSMUSG00000090667	16.82047	3.287885	0.84902	3.872565	0.000108	0.009257	Gm765	MDFIC2
ENSMUSG00000020251	121.1634	-1.07241	0.277143	-3.8695	0.000109	0.009274	Glt8d2	GLT8D2
ENSMUSG00000031928	214.0733	0.787714	0.20351	3.870632	0.000109	0.009274	Mre11a	MRE11
ENSMUSG00000069072	3692.318	-0.356	0.092243	-3.85939	0.000114	0.009615	Slc7a14	SLC7A14
ENSMUSG00000057110	403.1522	0.43024	0.111651	3.853438	0.000116	0.009793	Cntrl	CNTRL
ENSMUSG00000045268	39.26273	-2.59931	0.674738	-3.85232	0.000117	0.009793	Zfp691	ZNF691
ENSMUSG00000000555	18.95611	2.486112	0.645801	3.849658	0.000118	0.009835	Itga5	ITGA5
ENSMUSG00000030774	4513.545	-0.311	0.080805	-3.84874	0.000119	0.009835	Pak1	PAK1
ENSMUSG00000041180	1106.558	-0.60277	0.15668	-3.84715	0.00012	0.009848	Hectd2	HECTD2
ENSMUSG00000032332	214.0275	0.7688	0.200169	3.840761	0.000123	0.010056	Col12a1	COL12A1
ENSMUSG000000110924	20.19049	1.789686	0.468551	3.819616	0.000134	0.010839	Gm40518	
ENSMUSG00000020577	3246.709	-0.34547	0.09047	-3.81862	0.000134	0.010839	Tspan13	TSPAN13
ENSMUSG00000024298	3595.715	-0.37371	0.097866	-3.81854	0.000134	0.010839	Zfp871	ZNF433
ENSMUSG00000052430	626.5932	0.466549	0.122359	3.81296	0.000137	0.011031	Bmpr1b	BMPR1B
ENSMUSG00000038406	562.1296	1.206271	0.317127	3.803741	0.000143	0.011393	Scaf1	SCAF1
ENSMUSG00000033965	957.5411	0.418762	0.110229	3.799013	0.000145	0.011555	Slc16a2	SLC16A2
ENSMUSG00000055540	905.9178	-0.69114	0.182267	-3.7919	0.00015	0.011832	Epha6	EPHA6

ENSMUSG00000062202	729.8205	-0.52108	0.137797	-3.7815	0.000156	0.012277	Btbd9	BTBD9
ENSMUSG00000040407	3008.558	0.328177	0.086932	3.775084	0.00016	0.01244	Akap9	AKAP9
ENSMUSG00000025533	57.95648	1.750043	0.463604	3.77487	0.00016	0.01244	Asl	ASL
ENSMUSG00000047230	170.9864	0.917886	0.243178	3.77455	0.00016	0.01244	Cldn2	CLDN2
ENSMUSG00000085396	1317.305	0.333537	0.088454	3.770731	0.000163	0.012511	Firre	
ENSMUSG00000030314	165.0231	0.581558	0.154213	3.77114	0.000163	0.012511	Atg7	ATG7
ENSMUSG00000042246	216.8603	-0.53052	0.140776	-3.7685	0.000164	0.012562	Tmc7	TMC7
ENSMUSG00000062098	3433.524	-0.42818	0.113922	-3.75852	0.000171	0.013012	Btbd3	BTBD3
ENSMUSG00000061882	146.28	1.070334	0.284942	3.756321	0.000172	0.013065	Ccdc62	CCDC62
ENSMUSG00000013698	4214.137	0.303386	0.080799	3.75481	0.000173	0.013082	Pea15a	PEA15
ENSMUSG00000019370	7473.319	-0.26718	0.071185	-3.75329	0.000175	0.013099	Calm3	CALM3
ENSMUSG00000021983	1332.026	0.318077	0.084773	3.752106	0.000175	0.0131	Atp8a2	ATP8A2
ENSMUSG00000064329	2987.252	-0.30974	0.082862	-3.73807	0.000185	0.013789	Scn1a	SCN1A
ENSMUSG00000026841	284.7975	-0.88694	0.238023	-3.72627	0.000194	0.014384	Fibcd1	FIBCD1
ENSMUSG00000000794	606.3457	0.328689	0.088291	3.722806	0.000197	0.014515	Kcnn3	KCNN3
ENSMUSG00000060780	767.1811	-0.42499	0.114301	-3.71815	0.000201	0.014708	Lrrtm1	LRRTM1
ENSMUSG00000045817	394.0865	0.424649	0.11424	3.717152	0.000201	0.014708	Zfp36l2	ZFP36L2
ENSMUSG00000041468	259.0654	-0.51556	0.138947	-3.71044	0.000207	0.014967	Gpr12	GPR12
ENSMUSG00000042105	2120.16	-0.30617	0.082497	-3.71132	0.000206	0.014967	Inpp5f	INPP5F
ENSMUSG00000091264	3658.393	-0.28638	0.07729	-3.70526	0.000211	0.015207	Smim13	SMIM13
ENSMUSG00000031133	533.1943	0.633315	0.171191	3.69947	0.000216	0.015458	Arhgef6	ARHGEF6
ENSMUSG00000021236	463.9462	-0.39841	0.107735	-3.6981	0.000217	0.015458	Entpd5	ENTPD5
ENSMUSG00000050663	824.6048	-0.70085	0.189536	-3.69771	0.000218	0.015458	Trhde	TRHDE
ENSMUSG00000085438	13335.54	-0.38757	0.104858	-3.69614	0.000219	0.015485	Oip5os1	
ENSMUSG00000031983	416.7217	0.419351	0.113533	3.693651	0.000221	0.015568	2310022B05Rik	C1orf198
ENSMUSG00000004031	638.2136	-0.45732	0.123969	-3.68896	0.000225	0.015789	Brinp2	BRINP2
ENSMUSG00000066621	799.3888	-0.33602	0.091172	-3.68557	0.000228	0.015931	Tecpr1	TECPR1
ENSMUSG00000016487	734.1595	0.395712	0.107469	3.682112	0.000231	0.016078	Ppfibp1	PPFIBP1
ENSMUSG00000027273	34854.06	-0.24368	0.066228	-3.67942	0.000234	0.016179	Snap25	SNAP25
ENSMUSG00000036972	63.4875	1.568957	0.428116	3.66479	0.000248	0.017058	Zic4	ZIC4
ENSMUSG00000048279	3100.929	-0.46735	0.128048	-3.64981	0.000262	0.018006	Sacs	SACS
ENSMUSG00000036510	1027.532	-0.44408	0.121711	-3.6486	0.000264	0.018015	Cdh8	CDH8
ENSMUSG00000040016	56.24767	0.92925	0.254936	3.645036	0.000267	0.018188	Ptger3	PTGER3
ENSMUSG00000007653	1502.555	-0.27347	0.075101	-3.64139	0.000271	0.01837	Gabrb2	GABRB2
ENSMUSG00000067879	3290.14	-0.47729	0.131354	-3.63366	0.000279	0.018849	Vxn	VXN
ENSMUSG00000068184	140.5741	-0.87001	0.239889	-3.62673	0.000287	0.019041	Ndufaf2	NDUFAF2
ENSMUSG000000034751	1087.075	-0.50933	0.140335	-3.62942	0.000284	0.019041	Mast4	MAST4
ENSMUSG00000021687	2894.728	-0.28196	0.077735	-3.62718	0.000287	0.019041	Scamp1	SCAMP1
ENSMUSG00000047712	811.8046	-0.52349	0.144342	-3.62674	0.000287	0.019041	Ust	UST
ENSMUSG00000102649	884.437	-0.50635	0.140107	-3.61403	0.000301	0.019916	Gm38021	
ENSMUSG00000019943	29585.84	-0.49889	0.138138	-3.61152	0.000304	0.020027	Atp2b1	ATP2B1
ENSMUSG00000029168	226.2122	0.601514	0.16664	3.609663	0.000307	0.020088	Dpysl5	DPYSL5

ENSMUSG00000032300	78.11955	-0.79982	0.222116	-3.60091	0.000317	0.020692	1700017B05Rik	C15orf39
ENSMUSG00000032244	2771.684	-0.29763	0.082854	-3.59229	0.000328	0.021302	Fem1b	FEM1B
ENSMUSG00000048978	2862.159	-0.30343	0.084582	-3.58744	0.000334	0.021483	Nrsn1	NRSN1
ENSMUSG00000040254	396.7138	0.736136	0.205212	3.587206	0.000334	0.021483	Sema3d	SEMA3D
ENSMUSG000000091002	250.7922	0.874304	0.243747	3.586929	0.000335	0.021483	Tcerg1l	TCERG1L
ENSMUSG00000022237	849.8179	-0.46386	0.129401	-3.58468	0.000337	0.021582	Ankrd33b	ANKRD33B
ENSMUSG00000024330	155.7627	-0.68045	0.189958	-3.58212	0.000341	0.021708	Col11a2	COL11A2
ENSMUSG00000074736	111.1349	-0.75991	0.212211	-3.58094	0.000342	0.021719	Syndig1	SYNDIG1
ENSMUSG00000031530	104.9703	-0.63092	0.176401	-3.57664	0.000348	0.021906	Dusp4	DUSP4
ENSMUSG00000019966	1019.493	0.400852	0.112047	3.577549	0.000347	0.021906	Kittl	KITLG
ENSMUSG00000038172	1483.513	-0.39662	0.110942	-3.57504	0.00035	0.021954	Ttc39b	TTC39B
ENSMUSG00000006304	4394.953	-0.40393	0.113059	-3.57271	0.000353	0.022064	Arpc2	ARPC2
ENSMUSG00000007594	733.207	-0.38915	0.108983	-3.5707	0.000356	0.022147	Hapln4	HAPLN4
ENSMUSG00000078868	195.5118	-0.6056	0.170028	-3.56178	0.000368	0.022634	Gm14412	
ENSMUSG00000044835	845.5743	-0.48766	0.136894	-3.56235	0.000368	0.022634	Ankrd45	AL139142.2
ENSMUSG000000042425	385.4917	0.727822	0.204235	3.563657	0.000366	0.022634	Frmpd3	FRMPD3
ENSMUSG00000074637	561.2316	0.443517	0.12455	3.560951	0.00037	0.022634	Sox2	SOX2
ENSMUSG00000030451	4917.441	-0.29777	0.083653	-3.55962	0.000371	0.022662	Herc2	HERC2
ENSMUSG00000034731	5709.691	-0.52905	0.14868	-3.55834	0.000373	0.022686	Dgkh	DGKH
ENSMUSG00000029330	554.6138	-0.42382	0.119503	-3.54652	0.00039	0.023639	Cds1	CDS1
ENSMUSG00000025289	80.04234	0.720012	0.203542	3.537417	0.000404	0.024376	Prdx4	PRDX4
ENSMUSG00000001761	123.2777	0.601887	0.170298	3.534313	0.000409	0.02448	Smo	SMO
ENSMUSG000000038305	261.9526	0.609972	0.172578	3.534472	0.000409	0.02448	Spats2l	SPATS2L
ENSMUSG00000024074	2575.2	-0.44724	0.12664	-3.53162	0.000413	0.024638	Crim1	CRIM1
ENSMUSG000000113921	303.6595	-0.48971	0.13882	-3.52765	0.000419	0.024775	Gm48054	
ENSMUSG00000045589	7818.043	-0.4011	0.113683	-3.52823	0.000418	0.024775	Frrs1l	FRRS1L
ENSMUSG00000023915	981.8126	-0.35946	0.10191	-3.52721	0.00042	0.024775	Tnfrsf21	TNFRSF21
ENSMUSG00000039145	4780.612	-0.51286	0.145554	-3.52352	0.000426	0.02498	Camk1d	CAMK1D
ENSMUSG00000033615	1780.806	-0.328	0.0931	-3.52308	0.000427	0.02498	Cplx1	CPLX1
ENSMUSG00000048142	927.8028	-0.3508	0.099609	-3.52179	0.000429	0.02501	Nat8l	NAT8L
ENSMUSG00000021943	89.66288	0.62156	0.176637	3.518848	0.000433	0.025197	Gdf10	GDF10
ENSMUSG00000022324	575.2531	-0.63278	0.180404	-3.50759	0.000452	0.026192	Matn2	MATN2
ENSMUSG00000054027	2036.266	-0.86989	0.248514	-3.50037	0.000465	0.026722	Nt5dc3	NT5DC3
ENSMUSG00000031548	146.5958	0.761483	0.217546	3.500335	0.000465	0.026722	Sfrp1	SFRP1
ENSMUSG00000004151	4456.46	0.614183	0.175613	3.497371	0.00047	0.026924	Etv1	ETV1
ENSMUSG00000000326	20158.31	-0.37296	0.106813	-3.49171	0.00048	0.027403	Comt	COMT
ENSMUSG00000002017	728.6829	-0.33245	0.095316	-3.48784	0.000487	0.027659	Fam98a	FAM98A
ENSMUSG00000074575	328.303	0.758942	0.217629	3.487323	0.000488	0.027659	Kcng1	KCNG1
ENSMUSG00000038119	288.9873	0.485986	0.139544	3.482669	0.000496	0.028045	Cdon	CDON
ENSMUSG00000067038	293.5048	0.427698	0.123105	3.474246	0.000512	0.028838	Rps12-ps3	
ENSMUSG00000034040	1469.154	-0.60107	0.173426	-3.46585	0.000529	0.02965	Galnt17	GALNT17
ENSMUSG00000070348	627.084	-0.5136	0.148586	-3.45662	0.000547	0.030576	Ccnd1	CCND1

ENSMUSG00000053930	2752.301	-0.86166	0.249438	-3.45441	0.000551	0.03072	Shisa6	SHISA6
ENSMUSG00000020152	6799.049	-0.30351	0.088156	-3.44288	0.000576	0.031855	Actr2	ACTR2
ENSMUSG00000047945	766.5602	0.382192	0.111014	3.442739	0.000576	0.031855	Marcksl1	MARCKSL1
ENSMUSG00000019997	164.903	0.803357	0.2335	3.440499	0.000581	0.031989	Ccn2	CCN2
ENSMUSG00000031870	489.9922	-0.4763	0.138469	-3.43974	0.000582	0.031989	Pgr	PGR
ENSMUSG000000102275	1377.165	-0.42115	0.122686	-3.43272	0.000598	0.032604	Gm37144	
ENSMUSG00000057278	66.80556	0.745835	0.217253	3.433024	0.000597	0.032604	Snrpg	SNRPG
ENSMUSG00000020590	2102.635	-0.29266	0.085305	-3.43074	0.000602	0.032732	Snx13	SNX13
ENSMUSG00000026904	5088.38	0.416575	0.121607	3.425598	0.000613	0.033245	Slc4a10	SLC4A10
ENSMUSG00000016624	54.43059	-1.3342	0.389692	-3.42372	0.000618	0.033362	Phf21b	PHF21B
ENSMUSG00000042082	1625.565	-0.22299	0.065207	-3.41969	0.000627	0.033426	Arsb	ARSB
ENSMUSG00000020936	1562.587	-0.56001	0.163766	-3.41955	0.000627	0.033426	Nmt1	NMT1
ENSMUSG00000021699	528.2772	-0.37538	0.109718	-3.42129	0.000623	0.033426	Pde4d	PDE4D
ENSMUSG00000024410	300.5488	0.509272	0.148859	3.421184	0.000623	0.033426	Rmc1	RMC1
ENSMUSG00000029313	332.1584	-0.47599	0.139636	-3.40877	0.000653	0.03466	Aff1	AFF1
ENSMUSG000000105008	11008.81	-0.3795	0.11139	-3.40689	0.000657	0.034783	Gm43652	
ENSMUSG00000031283	99.49518	0.664707	0.195259	3.40424	0.000663	0.034925	Chrdl1	CHRD1
ENSMUSG00000027593	157.074	0.714517	0.209906	3.403984	0.000664	0.034925	Raly	RALY
ENSMUSG00000038936	866.3862	0.368604	0.108326	3.402736	0.000667	0.034969	Sccpdh	SCCPDH
ENSMUSG00000056899	36.94234	2.075587	0.610633	3.399076	0.000676	0.03521	Immp2l	IMMP2L
ENSMUSG00000042359	1879.788	-0.43026	0.126571	-3.39934	0.000675	0.03521	Osbpl6	OSBPL6
ENSMUSG00000032913	664.1244	-0.55544	0.163471	-3.39778	0.000679	0.035262	Lrig2	LRIG2
ENSMUSG00000025001	148.9715	0.538397	0.158802	3.390363	0.000698	0.036002	Hells	HELLS
ENSMUSG00000075040	60.33288	-0.90722	0.267591	-3.39032	0.000698	0.036002	Zfp408	ZNF408
ENSMUSG00000038932	25.59617	1.423525	0.420405	3.386079	0.000709	0.036446	Tcfl5	TCFL5
ENSMUSG00000052479	244.0629	-0.49214	0.14551	-3.38216	0.000719	0.036851	A330008L17Rik	
ENSMUSG00000038175	280.9308	-0.50341	0.148934	-3.38006	0.000725	0.037015	Mylip	MYLIP
ENSMUSG00000027335	33.48479	-1.05386	0.312181	-3.37579	0.000736	0.037135	Adra1d	ADRA1D
ENSMUSG00000036356	402.9208	0.420258	0.124443	3.377107	0.000733	0.037135	Csgalnact1	CSGALNACT1
ENSMUSG00000014030	69.07157	5.258045	1.556794	3.377483	0.000732	0.037135	Pax5	PAX5
ENSMUSG00000037300	837.0643	-0.69249	0.205141	-3.37568	0.000736	0.037135	Ttc13	TTC13
ENSMUSG00000028222	3159.411	0.356169	0.105935	3.362155	0.000773	0.038806	Calb1	CALB1
ENSMUSG00000027165	400.5186	-0.56224	0.167243	-3.36181	0.000774	0.038806	B230118H07Rik	IFTAP
ENSMUSG00000061013	221.3469	-0.65739	0.195647	-3.36009	0.000779	0.038927	Mkx	MKX
ENSMUSG00000089736	35.49717	1.695989	0.505603	3.354386	0.000795	0.039615	Tgfb3l	TGFBR3L
ENSMUSG00000019194	841.3926	-0.38543	0.114962	-3.35267	0.0008	0.039653	Scn1b	SCN1B
ENSMUSG00000045871	52.10889	-1.05004	0.313221	-3.3524	0.000801	0.039653	Slitrk6	SLITRK6
ENSMUSG00000040181	247.6483	0.594072	0.177412	3.348543	0.000812	0.040085	Fmo1	FMO1
ENSMUSG00000074415	2303.903	0.340372	0.101719	3.3462	0.000819	0.040196	Mir100hg	
ENSMUSG00000037709	367.0271	-0.53006	0.158411	-3.34607	0.00082	0.040196	Fam13a	FAM13A
ENSMUSG00000036264	167.4033	-0.70436	0.210817	-3.34109	0.000835	0.040428	Fstl4	FSTL4
ENSMUSG00000039316	66.46839	0.6882	0.205839	3.343395	0.000828	0.040428	Rftn1	RFTN1

ENSMUSG00000037593	99.19512	0.704792	0.210849	3.34264	0.00083	0.040428	Rskr	RSKR
ENSMUSG00000046699	1662.708	-0.45528	0.136261	-3.34121	0.000834	0.040428	Slitrk4	SLITRK4
ENSMUSG00000074305	1777.148	-0.4504	0.135114	-3.33349	0.000858	0.041423	Peak1	PEAK1
ENSMUSG00000022978	85.84239	-0.61653	0.185016	-3.33228	0.000861	0.041478	Mis18a	MIS18A
ENSMUSG00000040354	223.6371	-0.50288	0.151034	-3.32961	0.00087	0.041753	Mars1	MARS1
ENSMUSG00000056856	480.1189	0.295641	0.088872	3.326585	0.000879	0.042082	Jakmip3	JAKMIP3
ENSMUSG00000040125	2368.016	-0.28912	0.086963	-3.32465	0.000885	0.042248	Gpr26	GPR26
ENSMUSG00000016024	114.2572	0.93952	0.282985	3.320033	0.0009	0.042826	Lbp	LBP
ENSMUSG00000038976	7307.804	-0.25443	0.07668	-3.318	0.000907	0.04301	Ppp1r9b	PPP1R9B
ENSMUSG00000056770	2258.854	0.672609	0.202803	3.316558	0.000911	0.043105	Setd3	SETD3
ENSMUSG00000047414	1590.96	-0.36846	0.111129	-3.31561	0.000914	0.043124	Flrt2	FLRT2
ENSMUSG00000027583	124.901	0.718452	0.216746	3.31471	0.000917	0.043136	Zbtb46	ZBTB46
ENSMUSG00000022300	936.3884	0.327484	0.098938	3.309997	0.000933	0.04374	Dcaf13	DCAF13
ENSMUSG00000027474	24.63041	1.678843	0.507403	3.308694	0.000937	0.043816	Ccm2l	CCM2L
ENSMUSG00000022261	772.9067	0.278794	0.084354	3.305048	0.00095	0.04426	Sdc2	SDC2
ENSMUSG00000028617	155.2012	0.680046	0.206328	3.295938	0.000981	0.045588	Lrrc42	LRRC42
ENSMUSG00000111202	386.242	-0.59832	0.182078	-3.28604	0.001016	0.046976	Gm48275	
ENSMUSG00000026235	6345.103	-0.4388	0.133541	-3.28587	0.001017	0.046976	Epha4	EPHA4
ENSMUSG00000032802	452.9027	-0.39056	0.11899	-3.28229	0.00103	0.047439	Srxn1	SRXN1
ENSMUSG00000028613	2422.33	0.480587	0.146638	3.277372	0.001048	0.048135	Lrp8	LRP8